

Report on

University Course Registration

Prepared for

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Course Name: Information System Design & Software Engineering Lab

Prepared by

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Summary of the Project

Our university course registration system is designed to make the process of registering for courses easier for students by offering an intuitive interface. It enables students to search for courses, register, and choose them according to their preferences. It also gives them the option to change their selections before deciding on a schedule. Time scheduling capabilities are built into the system to assist students in minimizing conflicts and making the most of their course load. All things considered, it improves the registration process' flexibility and efficiency, guaranteeing users a flawless and easy experience.

Background Study

We researched the history of open credit universities as part of our project. Our main objectives were to gain a grasp of the open credit system, course registration procedures, and faculty assignment processes. Several tools were used in our research, such as Google, ChatGPT, and conversations with friends enrolled in other open credit institutions. We gained the fundamental knowledge needed to plan and execute a successful course registration system from this study.

Feasibility Analysis

Technical Feasibility:

- The required technologies, such as databases, frameworks, and programming languages, are suitable for creating the UCRS and satisfy the project's requirements.
- In order to ensure data availability and integrity, the system can effectively manage and store university course data while also accounting for backup and disaster recovery protocols.
- To ensure the continuous operation and optimization of the system, the project can maintain and support the UCRS throughout its lifecycle, taking into account issues with software upgrades, user assistance, and system management.
- Sophisticated security features, such as encryption, authentication, and access controls, are implemented by the system to guard against unauthorized access and data breaches of sensitive information.

Operational Feasibility:

- With the appropriate training, the university administration can efficiently administer and use the new UCRS by becoming familiar with its features, procedures, and best practices.
- To help office staff utilize the new UCRS efficiently, the System sets up user assistance methods.
- The project guarantees that the finished UCRS is in line with the university's strategic aims and objectives.

- Budget and infrastructure needs are evaluated as part of the project's assessment of the resources needed to execute and maintain the finished UCRS.

Economic Feasibility:

- The project carries out a comprehensive cost-benefit analysis to weigh the projected advantages of the UCRS against the estimated costs of creating, implementing, and maintaining it.
- In order to meet the demands of the University, the project assesses potential approaches and solutions, taking into account aspects like functionality, affordability, and compatibility with current systems.
- By evaluating budgetary constraints, cost overruns, and unplanned expenses, the project evaluates the financial risks associated with UCRS implementation.
- To optimize the utilization of financial resources and optimize the economic benefits of the UCRS, the project employs cost-effective measures.

Requirement Analysis

Functional:

1. **User Registration:** This functionality will allow Administration and users to log in to their account to change and look for information and do activities.
2. **Course Registration:** The main thing of open credit system is selecting courses and there will be flexibility in selecting courses.
3. **Course Search and Selection:** Course search and selection is also important in open credit system. User will be able to find course by searching every way possible
4. **Data modification flexibility:** The administration will have access to modify data of all possible things.
5. **Scheduling:** The software will ensure whether a student's course selection is colliding with other course selection and ensure proper scheduling.

Non-Functional:

1. **Security:** By using an encryption method to protect user information, the UCRS will be able to safeguard database information.
2. **Maintainability:** This is crucial since it accounts for about 75% of project costs. The system will be built with ease of maintenance in mind, enabling future improvements, upgrades, and bug patches.
3. **Reliability:** Users will rely on this system since, in typical circumstances, it will function dependably and gracefully deal with mistakes or malfunctions without causing data loss or damage.

4. **Scalability:** Since most universities in our nation and around the world require a system similar to this one, scalability won't be a problem with our system as we scale it through both national and international universities.
5. **Flexibility:** UCRS will be able to adjust to various circumstances, settings, and requirements without necessitating a substantial modification to its code or structure, making this system versatile.
6. **Performance:** Due to its strong server support, UCRS will operate at an excellent level. User requests will be handled by the system quickly, with reasonable turnaround times for operations like finding a course, registering for one, processing, etc.

Risk Analysis

This project is made to provide service to the students who're studying in open credit universities. So, these types of projects should provide service 24/7 without hampering and it has some risks as well.

Qualitative Risk Analysis:

- **Identification:** Possible problems that may occur while using HTML, CSS, JavaScript, PHP & MySQL.
- **Scalability Concerns:** It's important to provide all users a fluent user experience and for that proper database systems and infrastructure is needed.
- **Mitigation:** Implementing proper encryption system to ensure data safety.
- **Integration Complexity:** Our project has hard queries like time and course management and for that we have to use hard queries that may be difficult implementing in PHP.
- **System Downtime:** Sudden maintenance and unavoidable issues may cause system down and affect users.
- **Data Security & Privacy:** Occurrences like data breach and unauthorized access may occur in lack of proper security system.

Quantitative Risk Analysis:

- **Cost Overruns:** Facing complex things while undergoing the process may increase the cost.
- **Schedule Delays:** Possibilities of the project completion date may exceed due to complexity.

- **Data Collection:** We have to analyze the industry benchmarks as well as precautions about risks and their effects on the project.
- **Fault Tree Analysis:** Such things may hamper the database or PHP sessions or exception handling may not workable for such cases. The Fault Tree Analysis will show those elements that may hamper the project.

Challenges

- **Implementing Features:** Features like user authentication, course searching, time scheduling etc. may get hard while implementing as a lot of platforms like HTML, CSS, PHP are combining in same place.
- **Scalability:** Database may not perform properly if user activity gets much more than usual specially when result comes out.
- **Data Security:** We have to encrypt Admin/User panel data and passwords to ensure maximum security. So proper and strong encryption system is needed so that outsiders don't breach the data easily and get sensitive information and destroy as well.
- **Optimization issues:** If there are a lot of course, it may take a lot of time to get the outcome and it may cause performance drop.
- **Database Designing:** University course registration will need a relational database and so we need to ensure that all relations are working properly so that every user gets their own data.

In summary, University course registration is a challenging process but will be beneficial for the open course universities.

Project Goals

Our project's objective is to create an open credit model-appropriate, flexible, and effective university course registration system. Our goal is to establish a user-friendly platform that improves students' course choosing experience by utilizing insights obtained by analyzing the workings of open credit colleges, including course registration procedures and faculty assignments. The entire registration process will be streamlined by this system's seamless capabilities for managing schedules, changing selections, and registering courses.

Used Technologies for the project

The University Course Registration system utilizes a combination of frontend and backend technologies to deliver a robust and interactive web application.

Frontend Technologies:

- **HTML:** Structuring the content and layout of web pages.

- CSS: Styling HTML elements for visual presentation and user experience.
- JavaScript: Client-side scripting for interactive elements, form validation, and dynamic content updates.

Backend Technologies:

- PHP: Server-side scripting language for backend logic and server communication.
- MySQL: Relational database management system (RDBMS) for storing and managing data related to users, courses, and registrations.

Development Tools and Platforms:

- Visual Studio Code: Integrated development environment (IDE) for coding and debugging HTML, CSS, JavaScript, and PHP.
- XAMPP: Cross-platform web server solution stack (Apache, MySQL, PHP, Perl) for local development and testing.
- phpMyAdmin: Web-based tool for managing MySQL databases through a graphical user interface (GUI).

Additional Tools and Libraries:

- Bootstrap: Frontend framework for responsive and mobile-first design.
- AJAX: Asynchronous JavaScript and XML for making asynchronous HTTP requests.

These technologies and tools collectively enable the creation of a scalable, secure, and user-friendly University Course Registration system that meets both student and administrative requirements effectively.

Use Case Diagram

Here we'll show our software illustrations and its interactions using the use case diagram. It'll illustrate the features, functionalities and the service it'll provide person to person (Admin & Student).

The main subsystems of the project are Student Login, Student Dashboard, Admin Login and Register, Admin Dashboard, Admin Data Activity, Admin Data Modification. By this, it'll portrait a clear vision of how we will interact with the software. Here, each and every use cases will play an important role by ensuring user experience, data security and operationality.

Subsystems:

1. Student Login Subsystem:

- Use Cases: Login.
- Actors: Student.
- Prerequisites: Only admin can make profile for students. Then Student will use their mail and password to login to their profile.
- Priority: High.
- Course of Events:
 - I. Login: Student enters email and password and they get accessed to their personal profile.

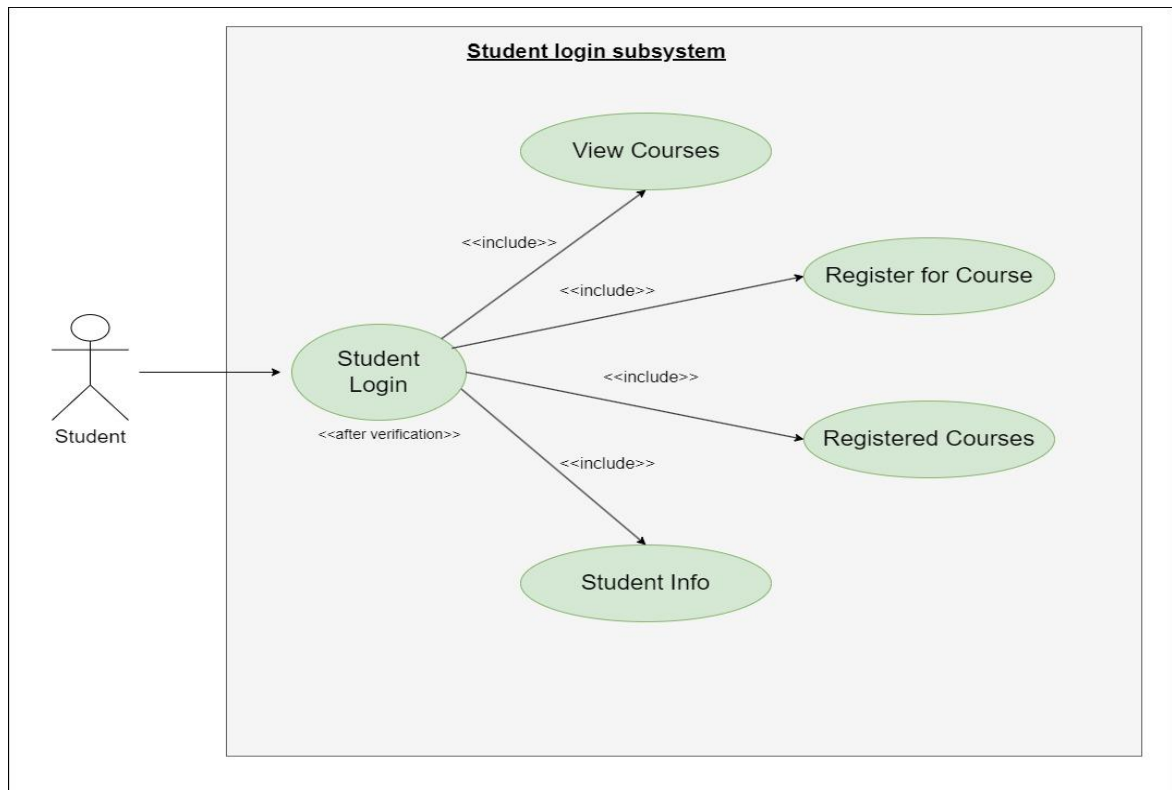


Figure 1 : Student Login Subsystem

2. Student Dashboard:

- Use Cases: View Courses, Register for Courses, View Profile.
- Actors: Student.
- Prerequisites: Student must log in to get access to dashboard.
- Priority: High.
- Course of Events:
 - I. View Courses: Student selects View courses to see the available courses.
 - II. Register for Courses: Student selects Register for courses, search for courses, register preferred courses.
 - III. View profile: Student selects View profile to see his/her information.

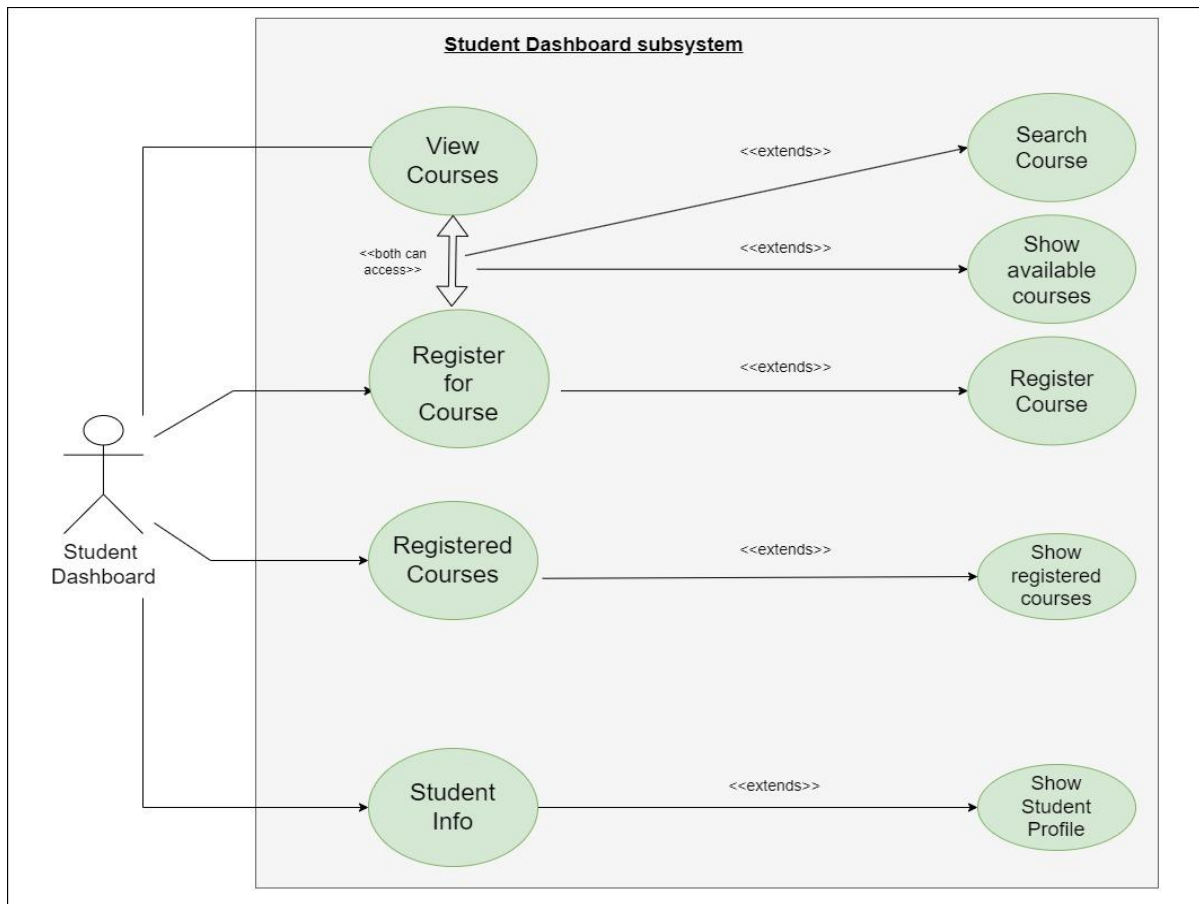


Figure 2: Student Dashboard Subsystem

3. Admin Login and Register:

- Use Cases: Register, Log In.
- Actors: Admin.
- Prerequisites: Admin must be sent an email to register for him/herself by providing username and password and furthermore he/she can log in using the username and password.
- Priority: High.
- Course of Events:
 - I. Register: An email will be sent to Admin and he/she provides username and password to get registered to the administrative sector.
 - II. Login: Admin enters username and password and they get accessed to the general administrative profile.

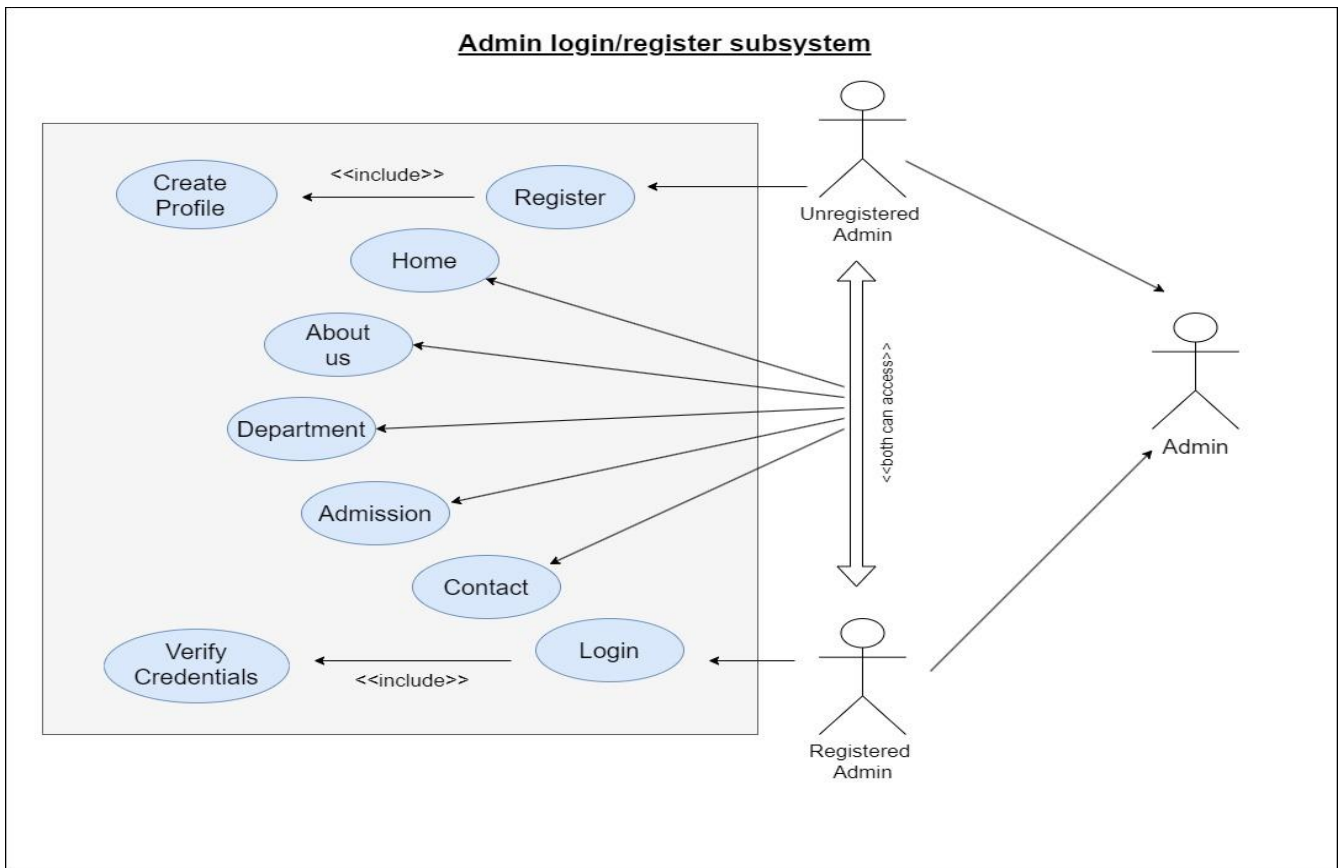


Figure 3: Admin login/register Subsystem

4. Admin Dashboard:

- Use Cases: Add Data, View Data.
- Actors: Admin
- Prerequisites: Admin must log in to the administrative account.
- Priority: High
- Course of Events:
 - I. Add Data: After logging in, Admin can add courses, department, instructor, student profile information to the database.
 - II. View Data: Admin can look up the information stored in database.

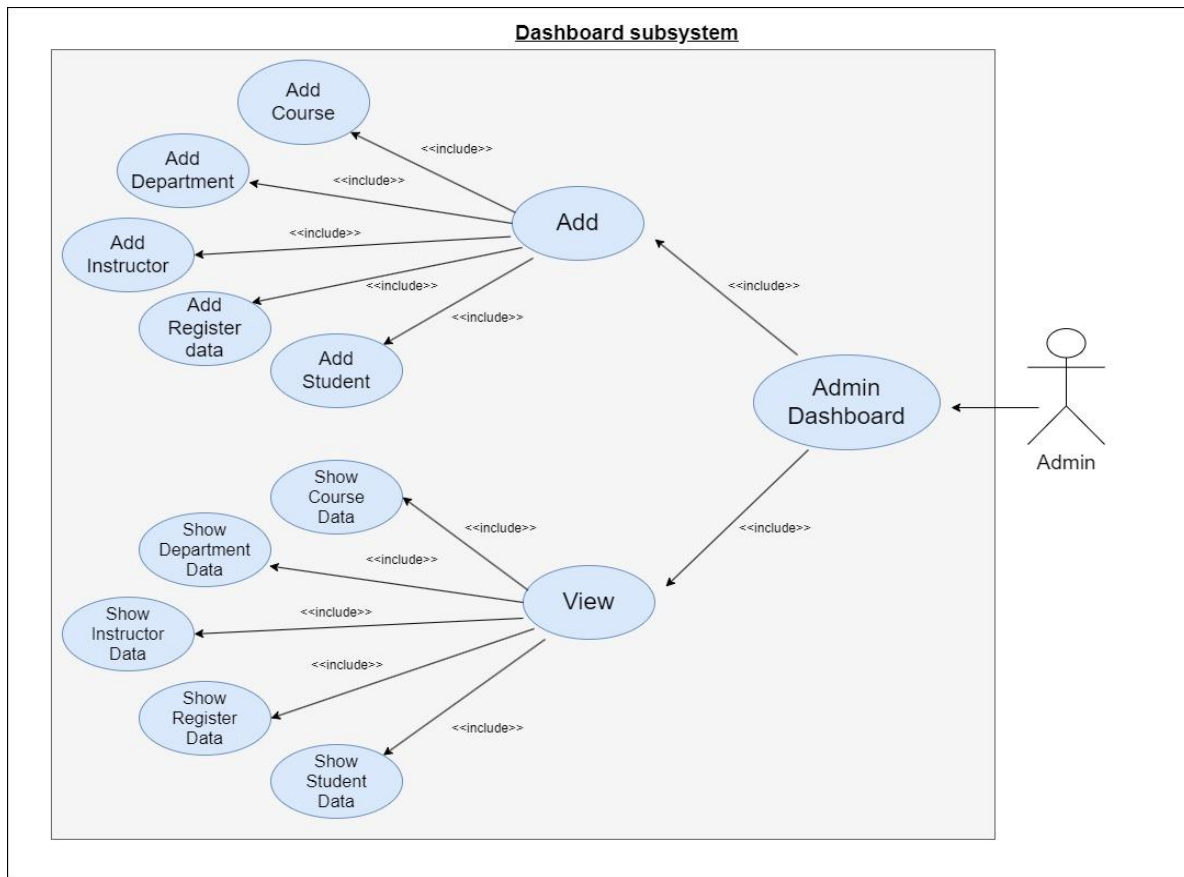


Figure 4: Admin Dashboard Subsystem

5. Admin Data Modification:

- Use Cases: Edit Data, Delete Data.
- Actors: Admin
- Prerequisites: Admin must log in to the administrative account and to his/her preferred area to change or delete data.
- Priority: Medium.
- Course of Events:
 - I. Edit Data: After logging in, Admin can edit courses, department, instructor, student profile information.
 - II. Delete Data: Admin can delete information from the database.

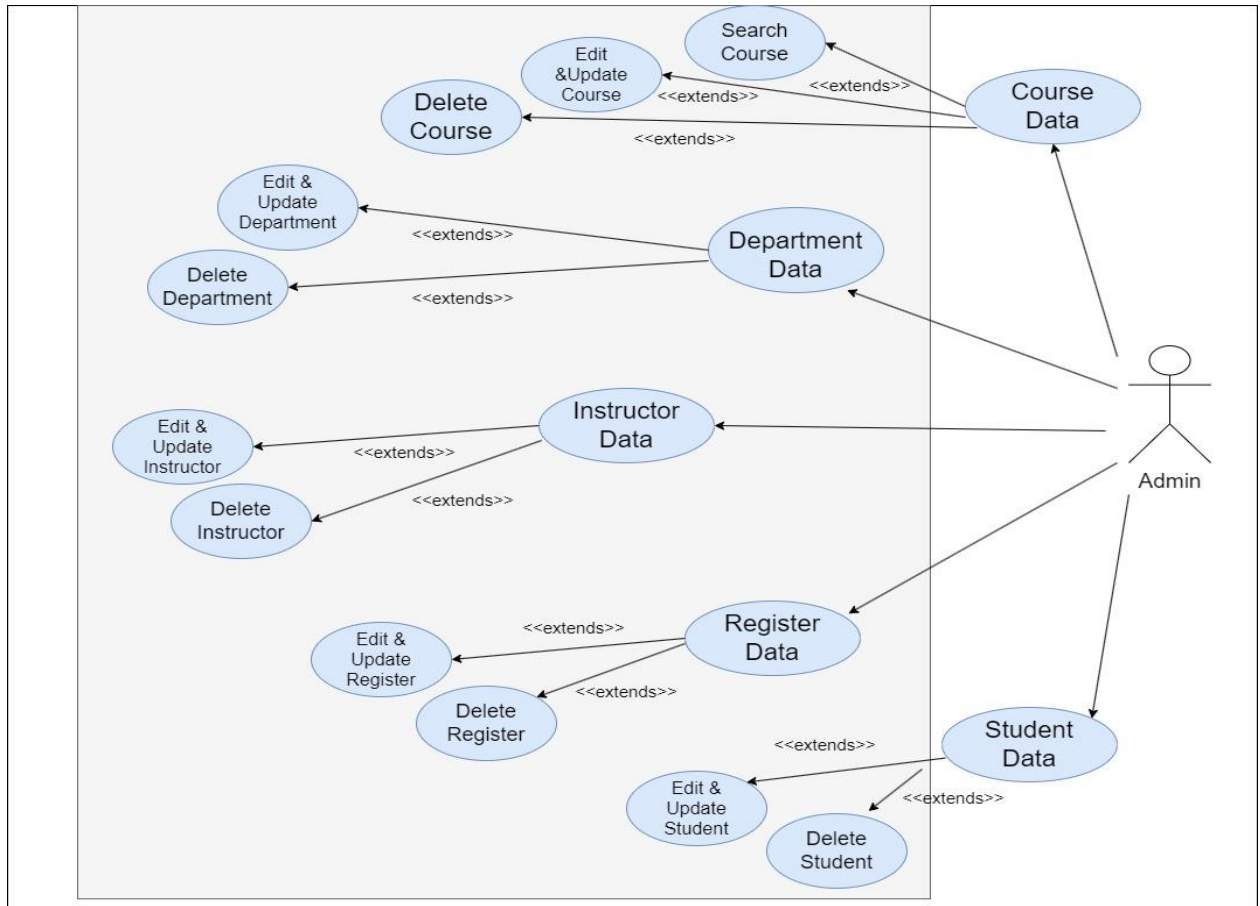


Figure 5: Admin Data Modification

Full Use Case Diagram:

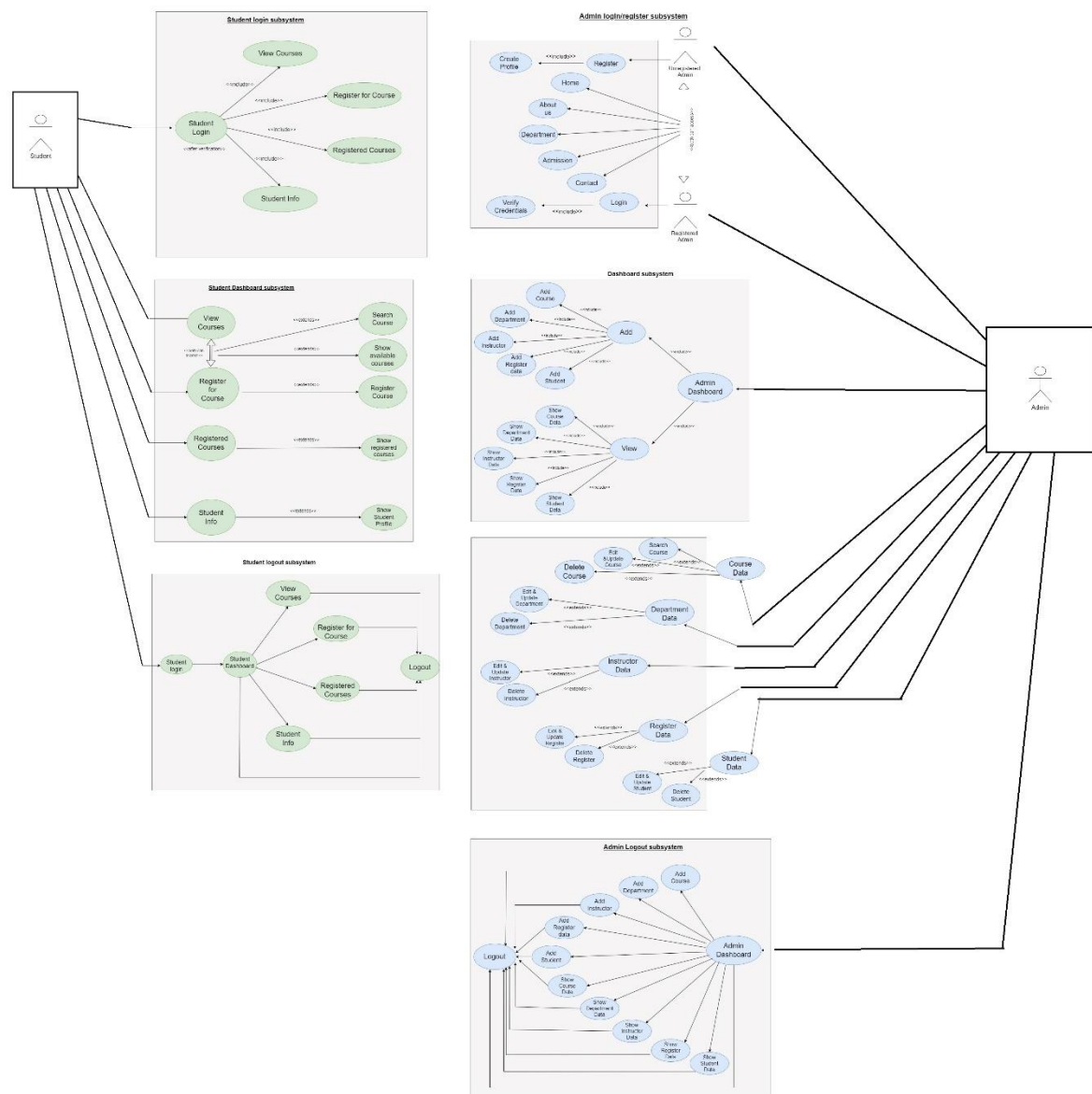


Figure 6: Full Use Case Diagram

Data Flow Diagram

A crucial tool for illustrating the flow of data through our university's course registration system is a data flow diagram (DFD). DFDs facilitate the understanding and communication of the design by assisting in the breakdown of the system's functionalities. DFDs are organized at several levels:

1. Levels of the Data Flow Diagram

1.1. Context Level Diagram

1. Provide a comprehensive picture of the system as a whole.
2. Demonstrates how the system communicates with other parties such as Admins and Students.

1.2. Diagram of Level 0

1. Divides the system's primary sub processes.
2. Highlights the key functional areas, including course registration, processing student data (register & login), course management, instructor management, dept. management, student management.

1.3. Diagrams at Level 1-n

1. Provides a thorough overview of every key functional area starting at Level 0.
2. Illustrates particular procedures and data flows in different domains.

2. Diagrams

2.1. Context Level Diagram

This diagram focuses on the main entities involved and the key processes, offering a simplified view of the system's boundaries and data flow. Here is a detailed explanation of the Context Level DFD for the University Course Registration System:

Entities:

Student: The end-user of the system who registers and enrolls in courses.

Admin: The system administrator who manages the overall system.

Main Process:

Course Registration System: The core application that processes all requests and actions.

Data Flows:

1. Students send requests to the Course Registration System for various services (registration, login, course enrollment).
2. The Course Registration System processes these requests and interacts with the respective databases to retrieve or store information.
3. The system sends notifications and responses back to the students, confirming actions or providing necessary updates.

4. Admins supply the system with essential data (creating courses, managing students) and receive notifications regarding system activities.

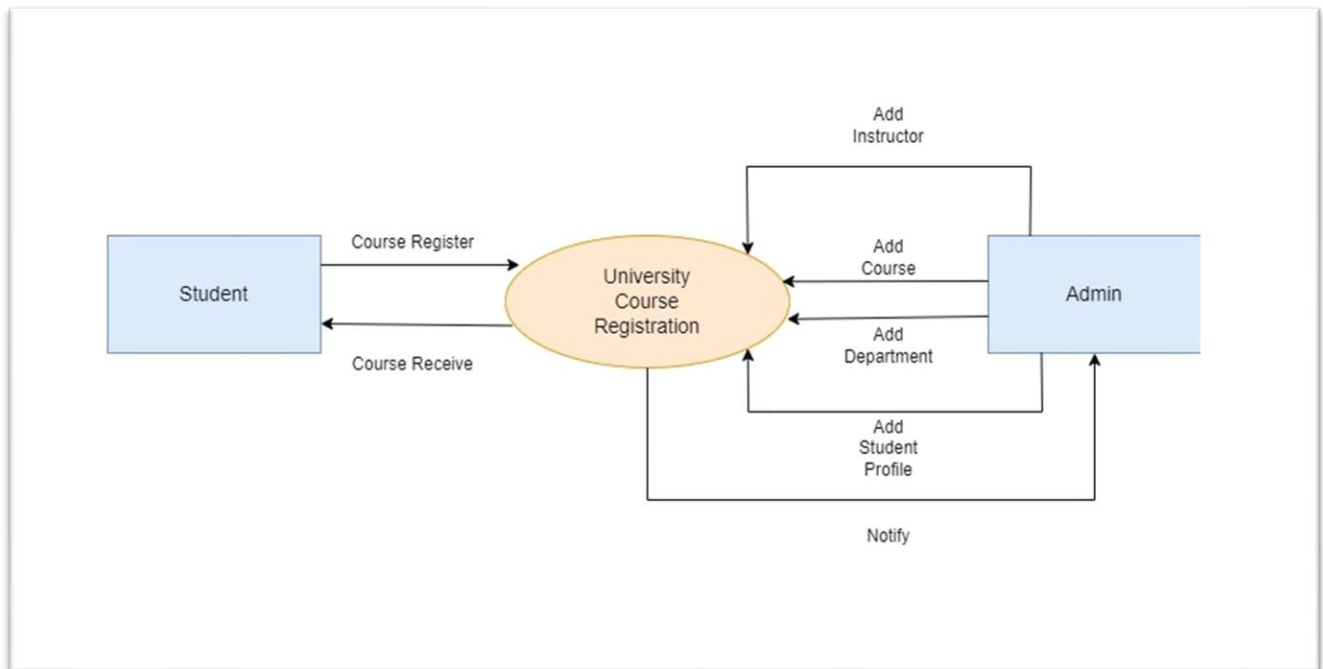


Figure 7: Context Level Diagram

2.2. Level 0 Diagram

The University Course Registration System's Level 0 Data Flow Diagram (DFD) offers a thorough rundown of all of the key functions and how they interact. It illustrates the data flow patterns between the user, the system, and external entities by dissecting the context-level diagram into seven main processes. The Level 0 DFD for the University Course Registration System is explained in full below:

2.2.1. User Registration

Entity: Admin

Process: Register

Data Stores: Student Database (DB)

Flow:

- 1.The admin submits registration information.
- 2.The system validates and stores student information in the Student DB.
- 3.The admin gives registration details (username & password) to student.

2.2.2. User Login

Entity: Student

Process: Login

Data Stores: Student Database (DB)

Flow:

- 1.The student submits login credentials.
- 2.The system verifies credentials against the Student DB.
- 3.Upon successful verification, the student's login status is updated.
- 4.Login confirmation is sent back to the student.

2.2.3. Course Registration

Entity: Student

Process: Enroll in Courses

Data Stores: Register Database, Course Database (DB)

Flow:

- 1.The student searches, selects and enrolls in courses.
- 2.The system updates the Register DB with the student's enrollment details.
- 3.Confirmation or rejection is sent back to the student.

2.2.4. Course Management

Entity: Admin

Process: Manage Courses

Data Stores: Course Database (DB)

Flow:

- 1.Admins create, search and manage course information.
- 2.Course details are stored in the Course DB.

2.2.5. Instructor Management

Entity: Admin

Process: Manage Instructors

Data Stores: Instructor Database (DB)

Flow:

- 1.Admins add, update, or view instructor information.
- 2.Instructor details are stored in the Instructor DB.

2.2.6. Department Management

Entity: Admin

Process: Manage Departments

Data Stores: Department Database (DB)

Flow:

- 1.Admins add, update, or view department information.

2.Department details are stored in the Department DB.

2.2.7. Student Management

Entity: Admin

Process: Manage Students

Data Stores: Student Database (DB)

Flow:

- 1.Admins add, update, or view student information.
- 2.Student details are stored in the Student DB.

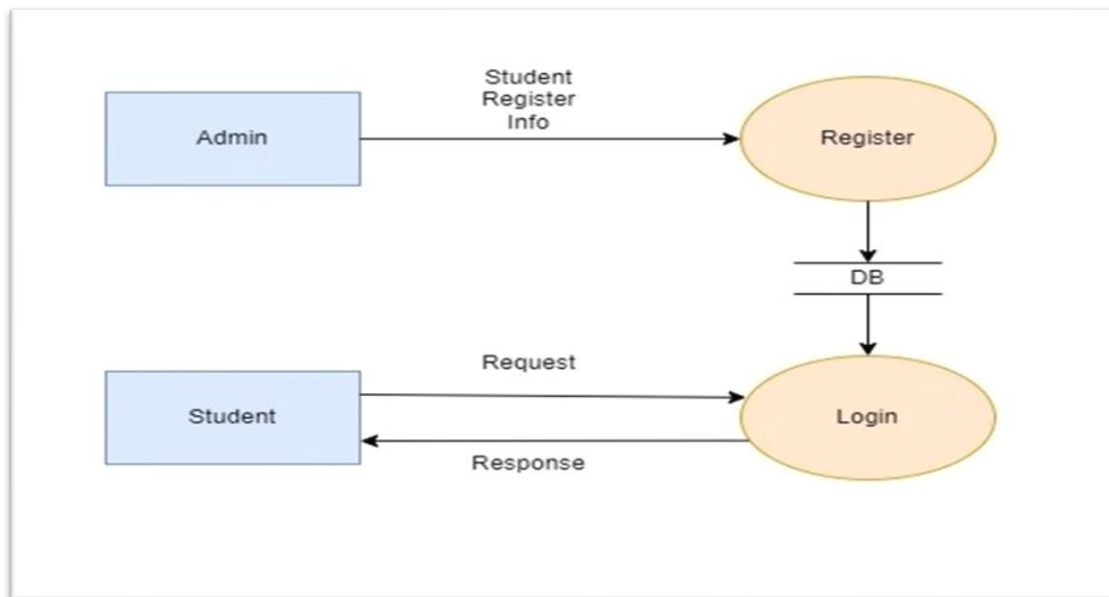


Figure 8: Level 0 Diagram (Part 1)

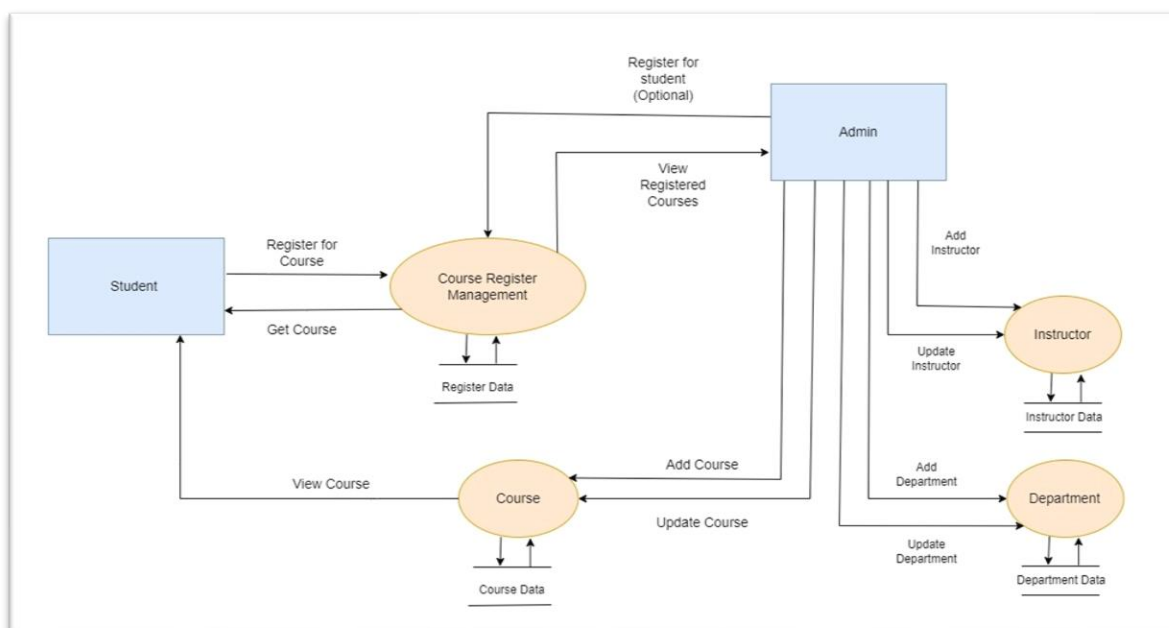


Figure 9: Level 0 Diagram (Part 2)

2.3. Level 1-n Diagram

Each major process described in the Context and Level 0 DFDs is broken down into more specific sub-processes in the University Course Registration System's Level 1 Data Flow Diagram (DFD). This level, which focuses on certain interactions and functions, offers a more detailed perspective of how data moves inside the system. A thorough explanation of each Level 1 component is provided below:

2.3.1. Student Registration

Process: Register

1.Submit Registration Information:

- Entity: Admin
- Data Store: Student Register Info (Temporary)
- Flow: The Admin submits students' registration details (username & password) which are stored temporarily for validation.

2.Validate Registration Information:

- Entity: System
- Data Store: Student Register Info (Temporary), Student Database (DB)
- Flow: The system validates the temporary registration information against existing records in the Student Database.

3.Store Student Information:

- Entity: System
- Data Store: Student Database (DB)
- Flow: Upon successful validation, the student information is stored in the Student Database. And admin gives it to student.

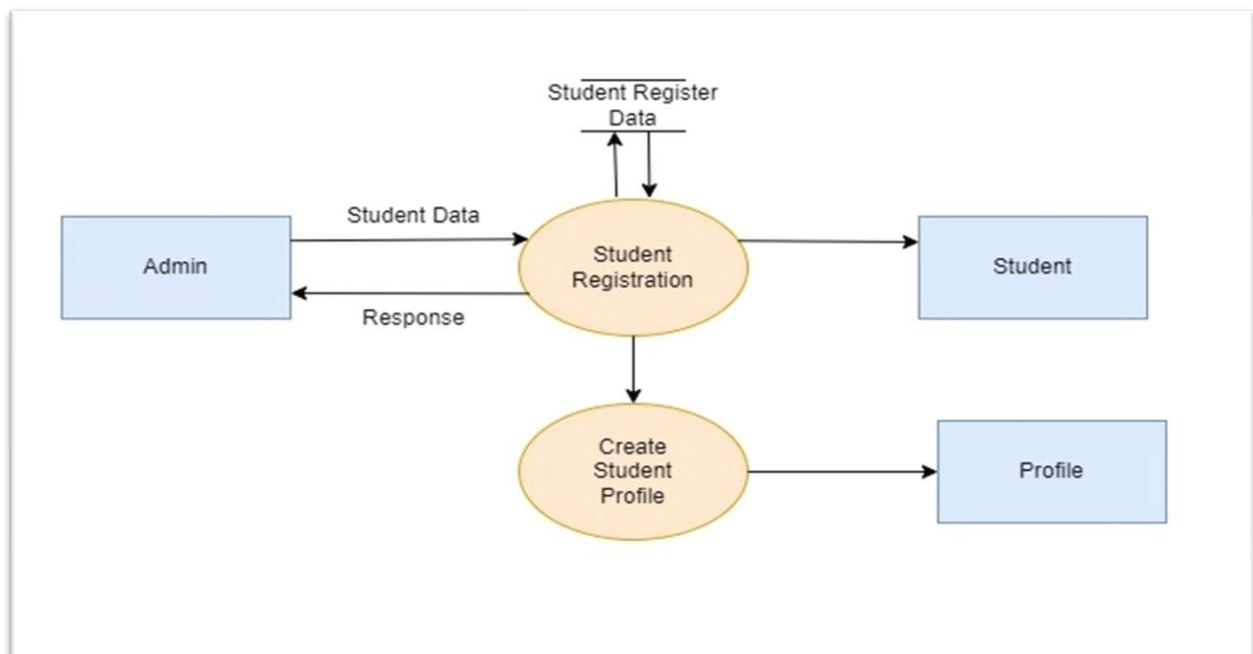


Figure 10: Level 1 Diagram (Student Registration)

2.3.2. Student Login

Process: Login

1.Submit Login Credentials:

- Entity: Student
- Data Store: Login Info (Temporary)
- Flow: The student submits their login credentials, which are stored temporarily for verification.

2.Verify Credentials:

- Entity: System
- Data Store: Student Database (DB)
- Flow: The system verifies the credentials against the Student Database.

3.Update Login Status:

- Entity: System
- Flow: Upon successful verification, the student's login status is updated.

4.Send Login Confirmation:

- Entity: System
- Flow: The system sends a login confirmation back to the student.

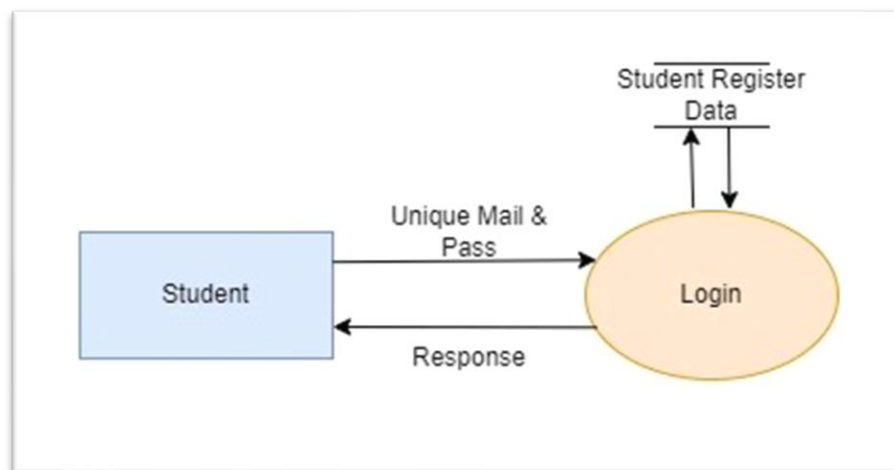


Figure 11: Level 1 Diagram (Student Login)

2.3.3. Register for courses

Process: Course Registration

1.Select Courses:

- Entity: Student
- Data Store: Register Info (Temporary)
- Flow: The student selects courses for enrollment, which are stored temporarily for validation.

2.Validate Registration:

- Entity: System
- Data Store: Register Info (Temporary), Course Database (DB)

- Flow: The system validates the register details against the Course Database for seat availability.

3.Store Register Details:

- Entity: System
- Data Store: Register Database
- Flow: Upon successful validation, the enrollment details are stored in the Register Database.

4.Send Confirmation:

- Entity: System
- Flow: The system sends an enrollment confirmation or rejection message back to the student.

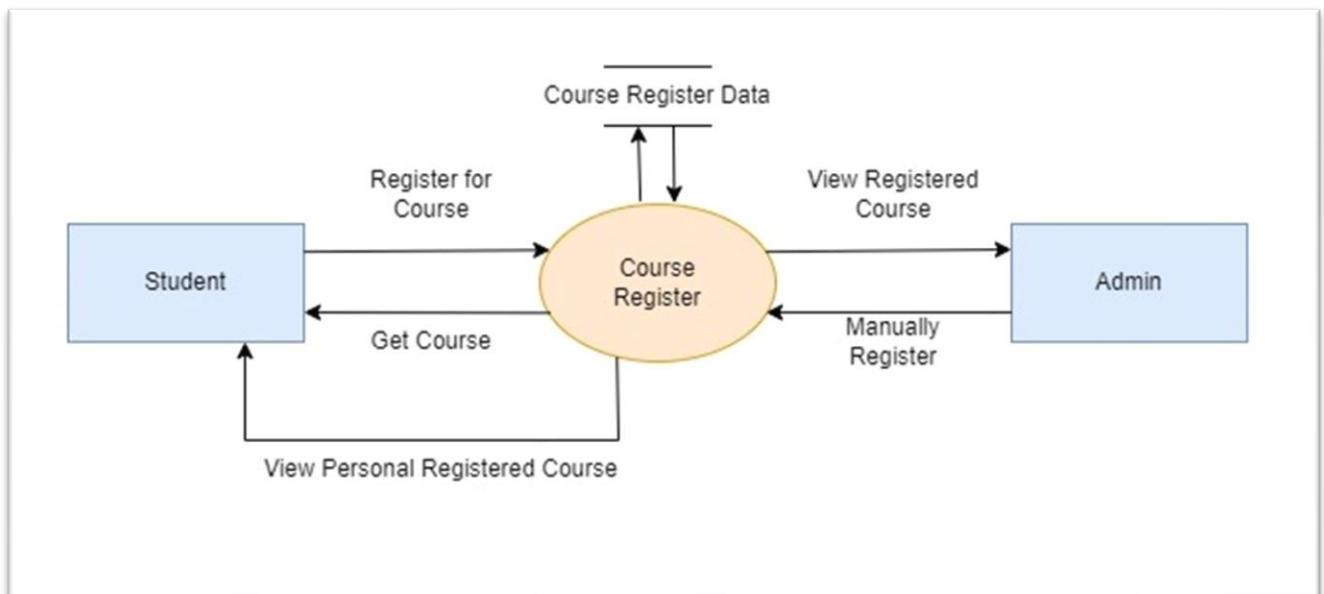


Figure 12: Level 1 Diagram (Course Register)

2.3.4. Course Management

Process: Manage Courses

1.Create Course:

- Entity: Admin
- Data Store: Course Info (Temporary)
- Flow: The admin creates course details, which are stored temporarily for validation.

2.Store Course Details:

- Entity: System
- Data Store: Course Database (DB)
- Flow: Validated course details are stored in the Course Database.

3.Update Course:

- Entity: Admin
- Data Store: Course Database (DB)

- Flow: Admins update course information, which is then stored in the Course Database.

4.View Course:

- Entity: Admin/Student
- Data Store: Course Database (DB)
- Flow: Admins and students view course information stored in the Course Database.

5.Search Course:

- Entity: Admin/Student
- Data Store: Course Database (DB)
- Flow: Admins and students search for specific course information in the Course Database based on various criteria.

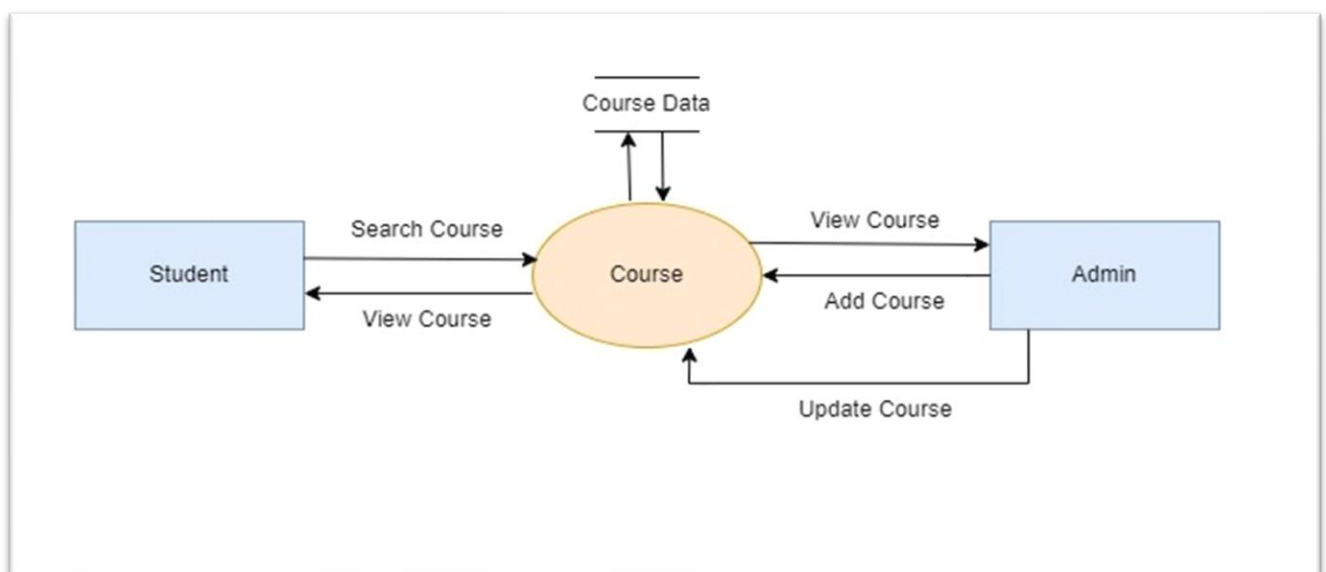


Figure 13: Level 1 Diagram (Course Info)

2.3.5. Instructor Management

Process: Manage Instructors

1.Add Instructor:

- Entity: Admin
- Data Store: Instructor Info (Temporary)
- Flow: The admin adds instructor details, which are stored temporarily for validation.

2.Store Instructor Details:

- Entity: System
- Data Store: Instructor Database (DB)
- Flow: Validated instructor details are stored in the Instructor Database.

3.Update Instructor:

- Entity: Admin
- Data Store: Instructor Database (DB)

- Flow: Admins update instructor information, which is then stored in the Instructor Database.

4.View Instructor:

- Entity: Admin/Student
- Data Store: Instructor Database (DB)
- Flow: Admins and students view instructor information stored in the Instructor Database.

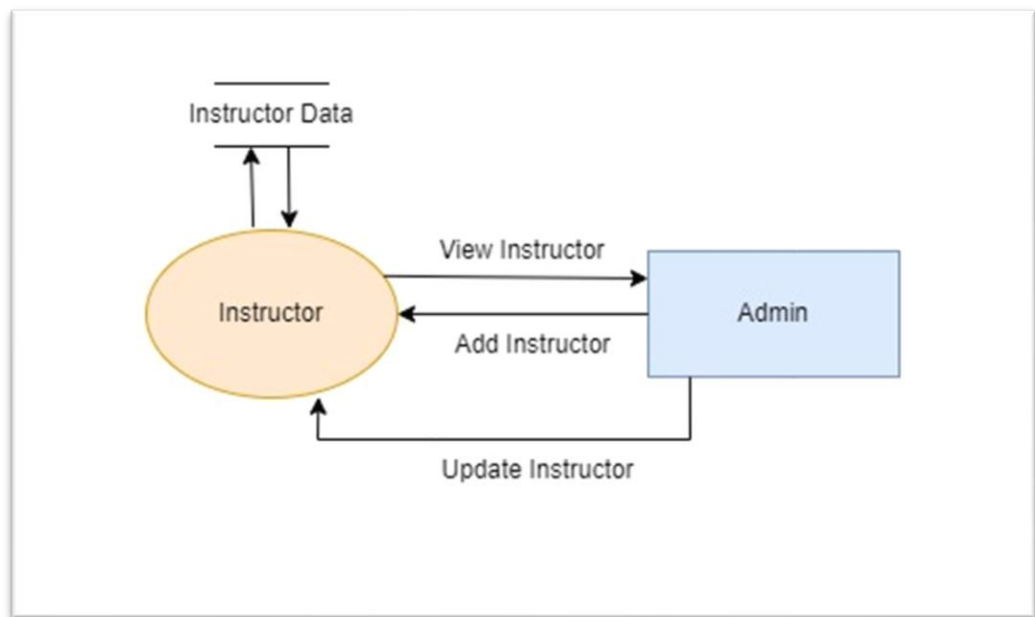


Figure 14: Level 1 diagram (Instructor Info)

2.3.6. Department Management

Process: Manage Departments

1.Add Department:

- Entity: Admin
- Data Store: Department Info (Temporary)
- Flow: The admin adds department details, which are stored temporarily for validation.

2.Store Department Details:

- Entity: System
- Data Store: Department Database (DB)
- Flow: Validated department details are stored in the Department Database.

3.Update Department:

- Entity: Admin
- Data Store: Department Database (DB)
- Flow: Admins update department information, which is then stored in the Department Database.

4.View Department:

- Entity: Admin/Student
- Data Store: Department Database (DB)
- Flow: Admins and students view department information stored in the Department Database.

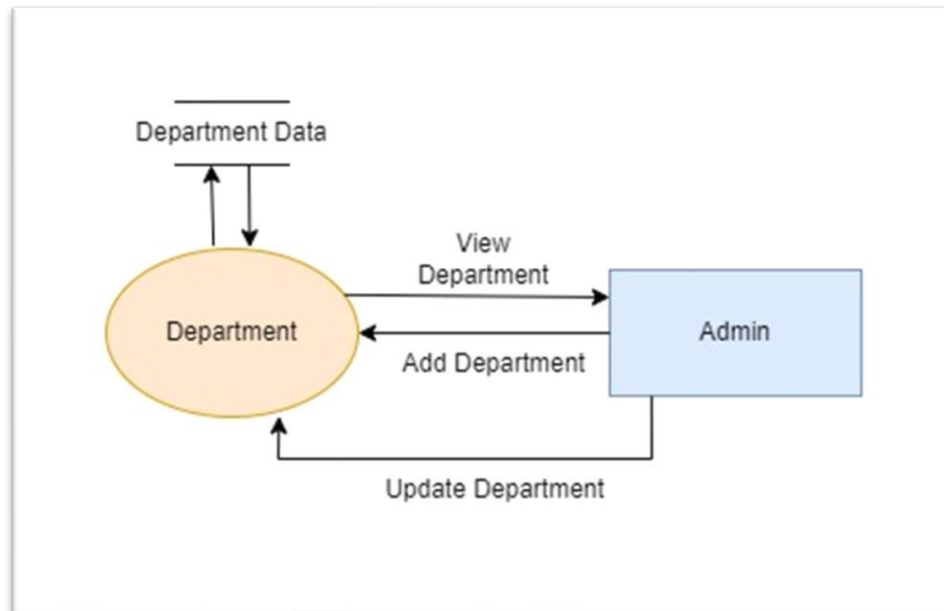


Figure 15: Level 1 Diagram (Department Info)

2.3.7. Student Management

Process: Manage Students

1.Add Student:

- Entity: Admin
- Data Store: Student Info (Temporary)
- Flow: The admin adds student details, which are stored temporarily for validation.

2.Store Student Details:

- Entity: System
- Data Store: Student Database (DB)
- Flow: Validated student details are stored in the Student Database.

3.Update Student:

- Entity: Admin
- Data Store: Student Database (DB)
- Flow: Admins update student information, which is then stored in the Student Database.

4.View Student:

- Entity: Admin

- Data Store: Student Database (DB)
- Flow: Admins view student information

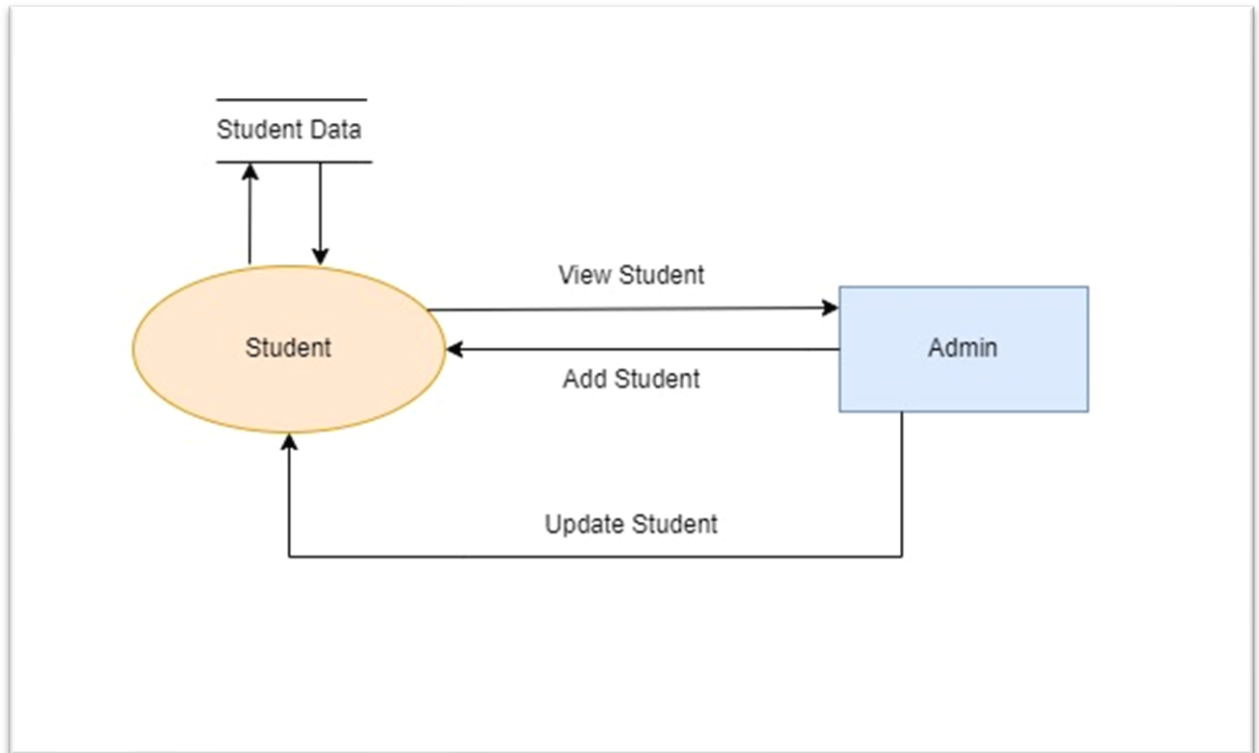


Figure 16: Level 1 Diagram (Student Info)

Entity Relation Diagram

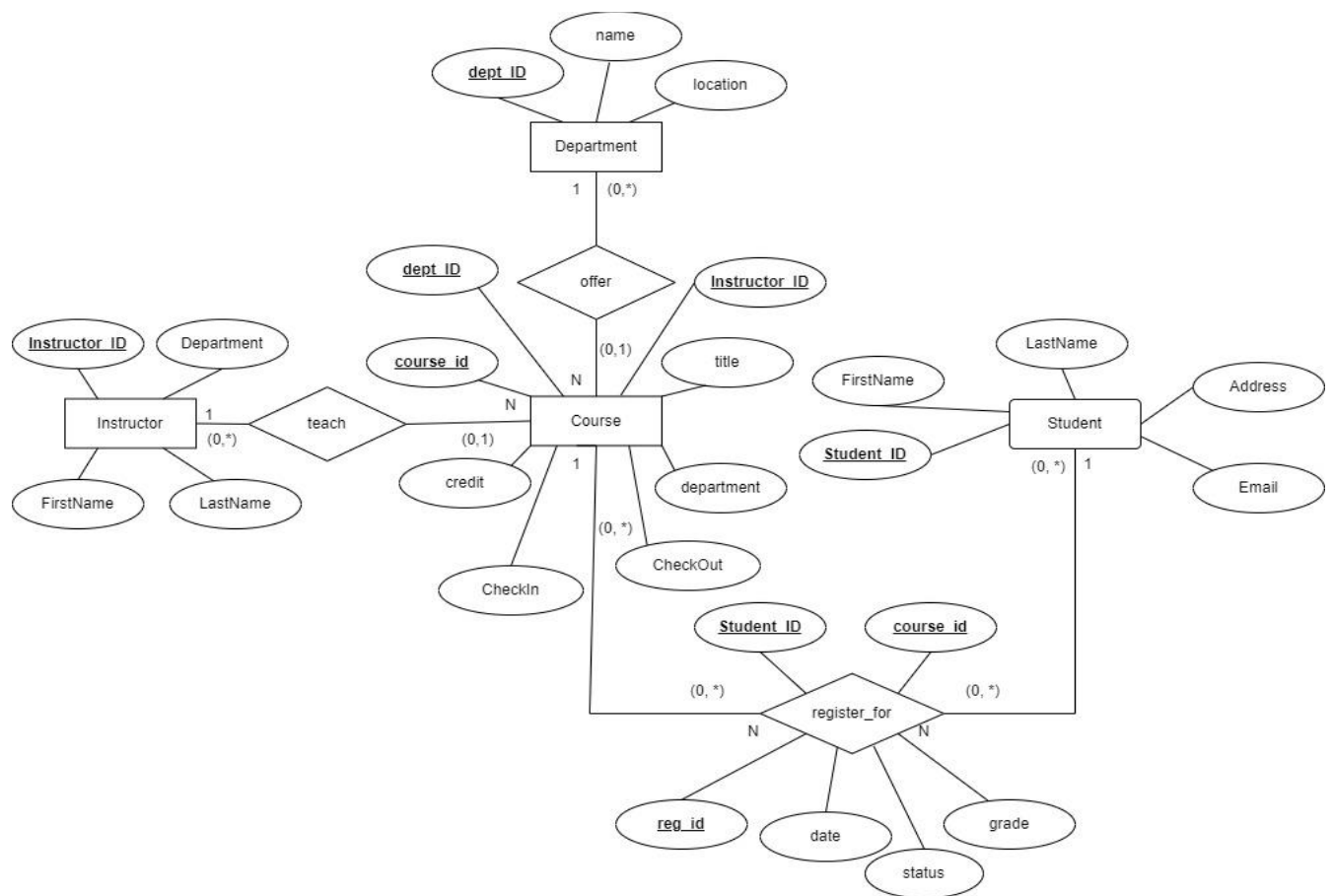


Figure 17: ERD of University Course Registration

Entities:

- Department**
 Attributes: dept_ID, name, location
- Instructor**
 Attributes: Instructor_ID, FirstName, LastName, Department
- Course**
 Attributes: course_id, title, credit, department, CheckIn, CheckOut
- Student**
 Attributes: Student_ID, FirstName, LastName, Address, Email
- Register_for**
 Attributes: reg_id, date, grade, status

Relationship:

In the ERD, the Department can offer multiple Courses, with each course being associated with only one department. This is represented by the relationship "offer," where the cardinality is (0,1) on the Department side and N on the Course side. Instructors are linked to departments with a one-to-many relationship, meaning an instructor belongs to only one department, but a department can have many instructors. This relationship is not explicitly named but could be interpreted as "works for" or "belongs to." Instructors and courses share a many-to-many relationship through the "teach" relationship, where an instructor can teach multiple courses, and a course can be taught by multiple instructors.

Students register for courses through the associative entity Register_for, which tracks registration details such as date, grade, and status. The relationship between Course and Student is many-to-many, where a student can register for many courses, and a course can have many students registered. The Register_for entity connects both students and courses, with a many-to-many cardinality on both sides, ensuring that the registration process is well-represented.

Project Features (Include descriptions and screenshots)

Home Page: The Hogwarts University Home Page, Hogwarts Castle welcomes visitors with buttons for courses, faculty, and admission, along with rotating images of key locations.



Figure: Home Page

About Us Page: The About Us Page highlights Hogwarts' history, its founders, faculty, and a timeline of the school's evolution.



Figure: About us Page

Dept. Page: The Department Selection Page features Hogwarts Castle with the university crest and title. A navigation bar links to key sections, and central buttons represent departments like CSE, EEE, and Arts.

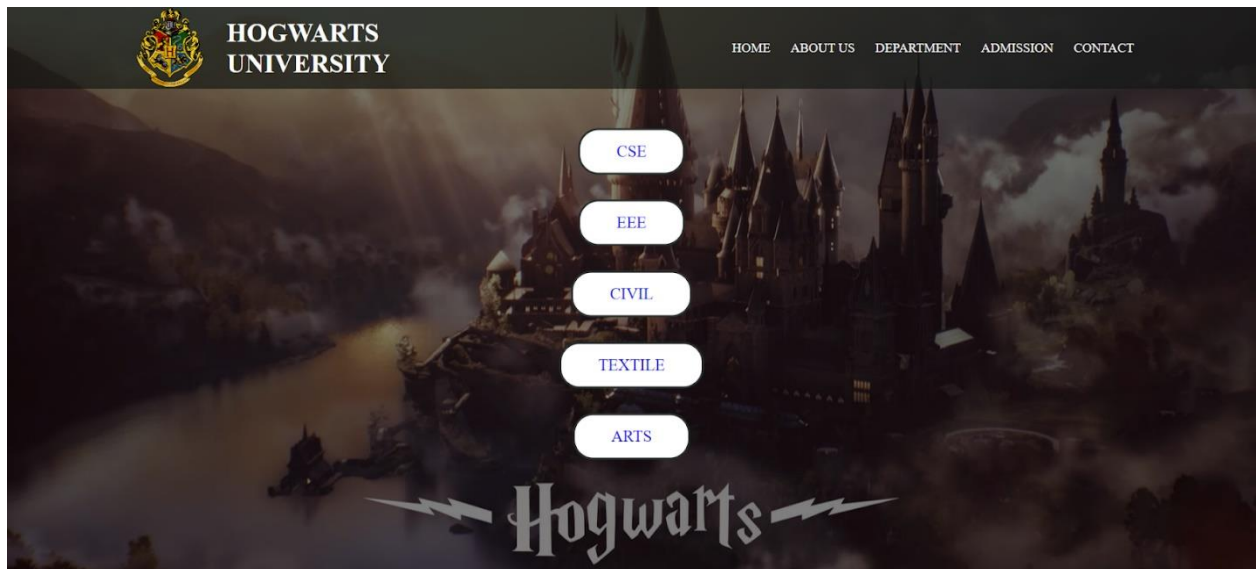


Figure: Dept. Page

Admission Page: The Admission Page provides details on the process, humorously noting owl-delivered letters, with a “Get Admission” button at the bottom.



Figure: Admission Page

Admin Login: The system has a secure admin login page where authorized personnel can access the management dashboard.

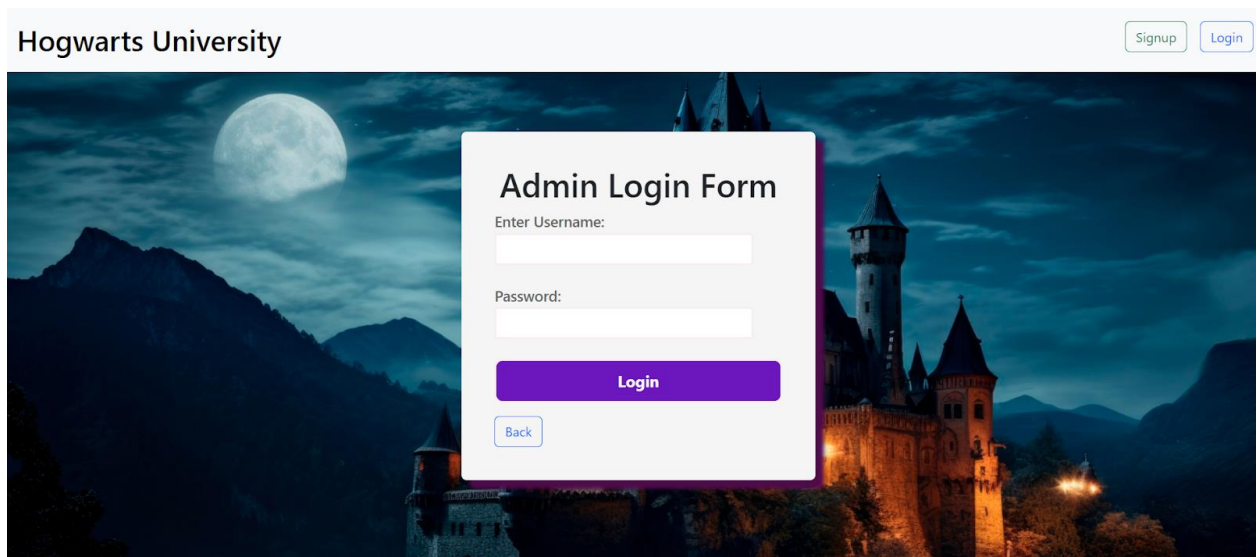


Figure: Admin Login Page

Admin Dashboard: A centralized dashboard provides a professional and user-friendly interface for admins to view statistics, manage entities, and access the various modules.

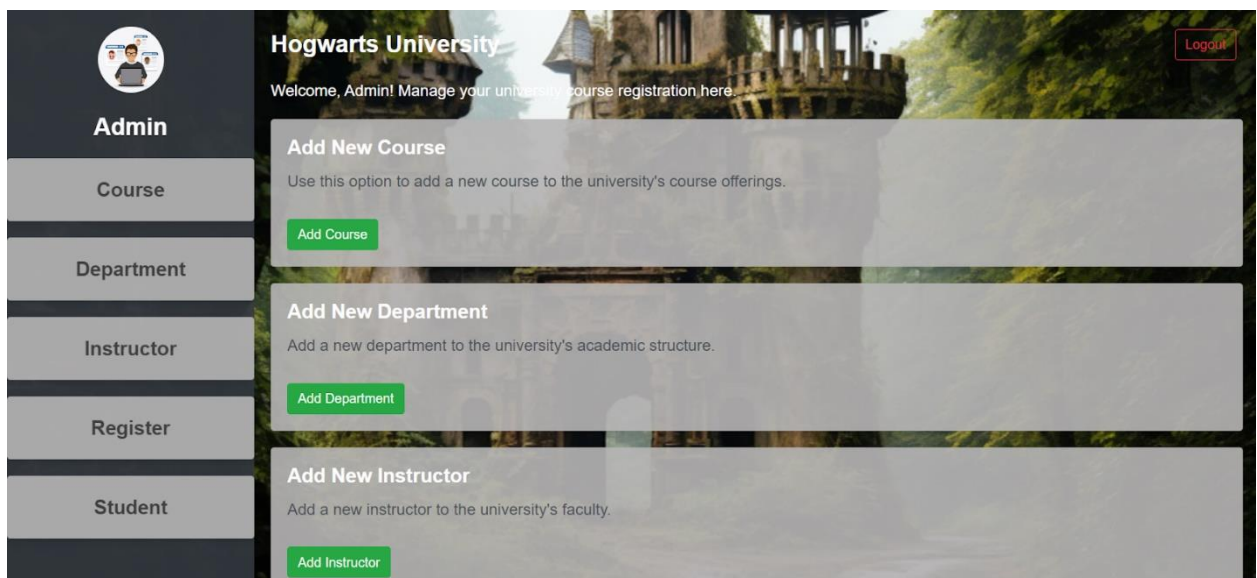
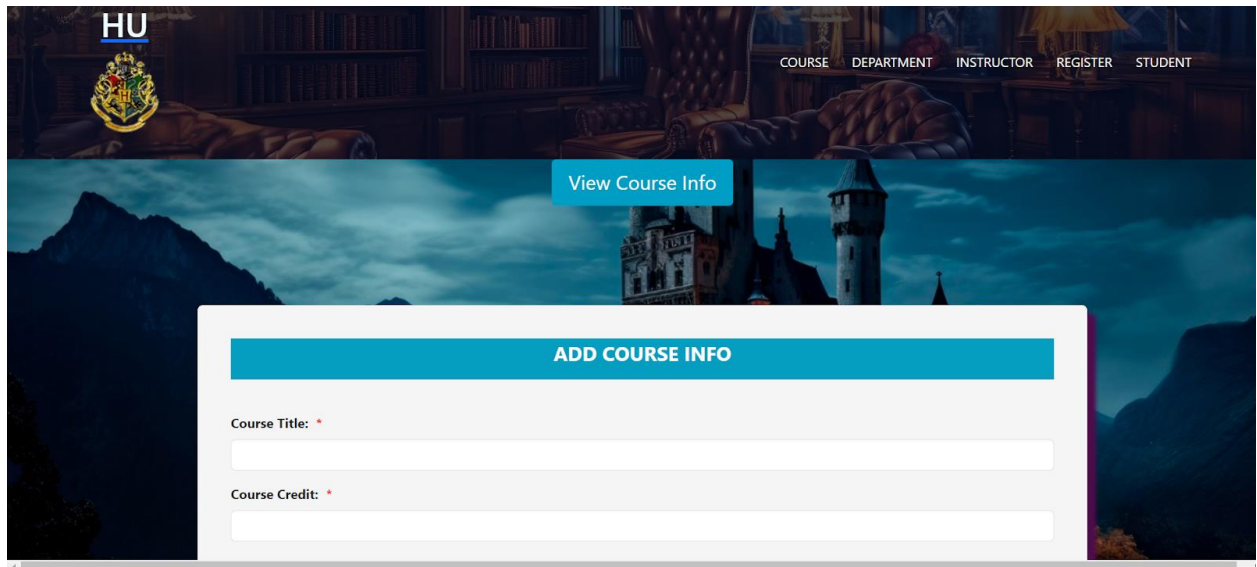


Figure: Admin Dashboard Page

Course Management (Course Add, Course Data):

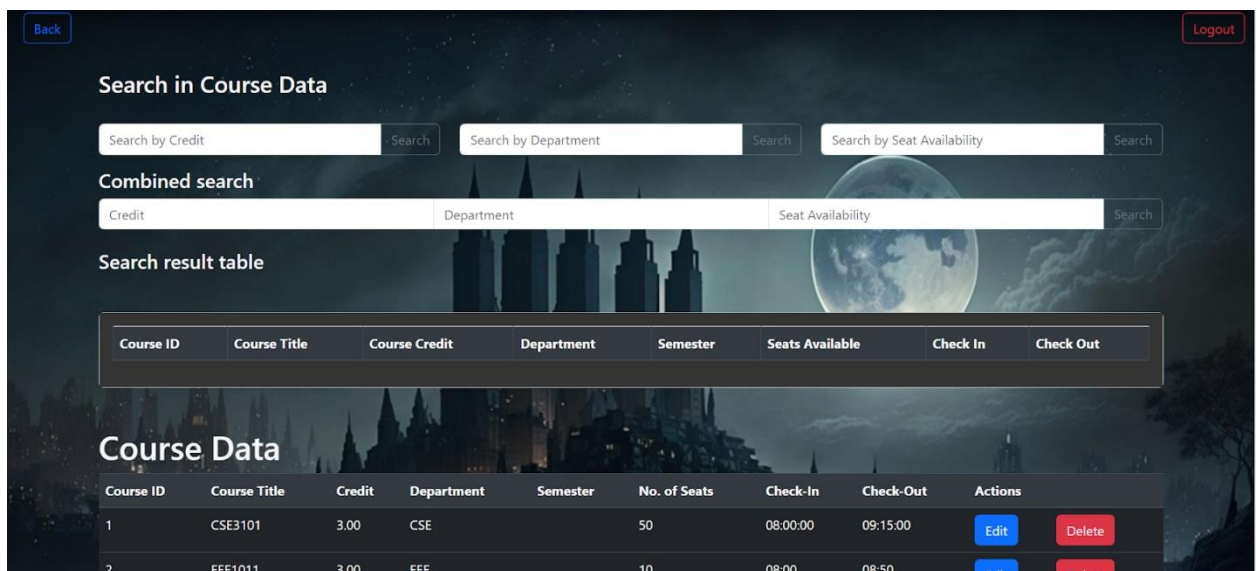
Administrators can add new courses and manage existing ones. The CourseAdd feature allows inputting course details such as name, description, credits, and department assignment.



The screenshot shows the 'ADD COURSE INFO' form on the HU website. The form is overlaid on a background image of a castle at night. The form has a blue header bar with the text 'ADD COURSE INFO'. Below the header, there are two input fields: 'Course Title: *' and 'Course Credit: *'. A 'View Course Info' button is visible above the form.

Figure: Course Add Page

The CourseData section offers a consolidated view of all courses, with options to edit or delete, search them.



The screenshot shows the 'Course Data' page with a search interface and a table of course data. The search interface includes three search boxes: 'Search by Credit', 'Search by Department', and 'Search by Seat Availability'. Below these is a 'Combined search' section with three input fields: 'Credit', 'Department', and 'Seat Availability'. The 'Search result table' is a table with the following columns: Course ID, Course Title, Course Credit, Department, Semester, Seats Available, Check In, and Check Out. The 'Course Data' table below it has the following columns: Course ID, Course Title, Credit, Department, Semester, No. of Seats, Check-In, Check-Out, and Actions. The Actions column contains 'Edit' and 'Delete' buttons for each course.

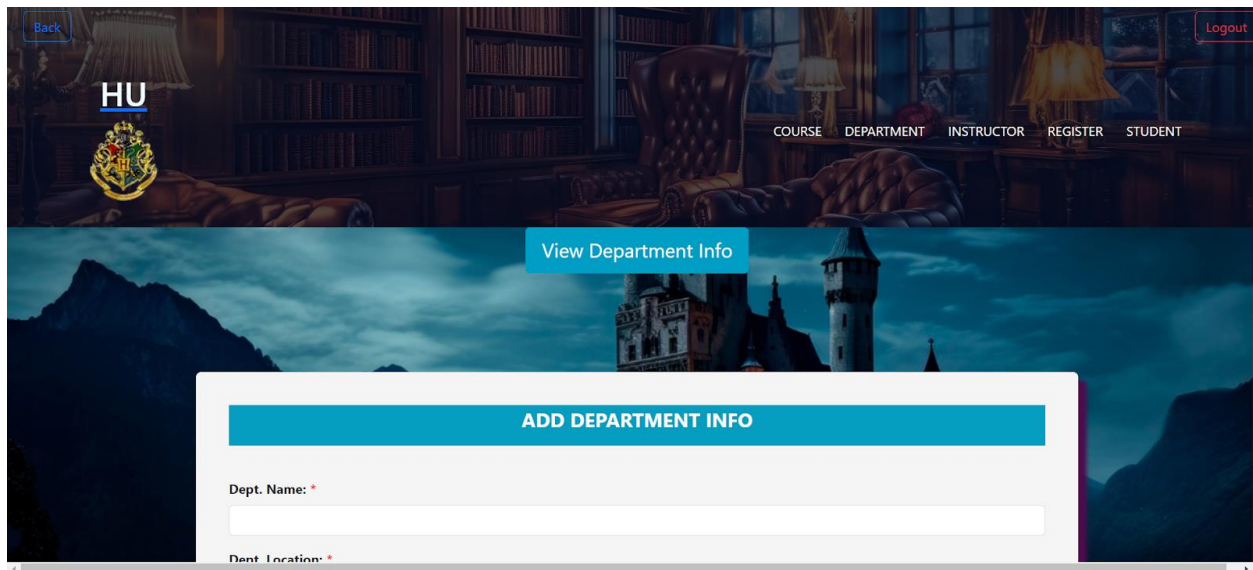
Course ID	Course Title	Credit	Department	Semester	Seats Available	Check In	Check Out
1	CSE3101	3.00	CSE		50	08:00:00	09:15:00
2	EEE1011	3.00	EEE		10	08:00	08:50

Course ID	Course Title	Credit	Department	Semester	No. of Seats	Check-In	Check-Out	Actions
1	CSE3101	3.00	CSE		50	08:00:00	09:15:00	Edit Delete
2	EEE1011	3.00	EEE		10	08:00	08:50	Edit Delete

Figure: Course Data Page

Department Management (DeptAdd, DeptData):

The DeptAdd feature enables the addition of academic departments, specifying details like department name, head of department, and contact information



Back Logout

COURSE DEPARTMENT INSTRUCTOR REGISTER STUDENT

View Department Info

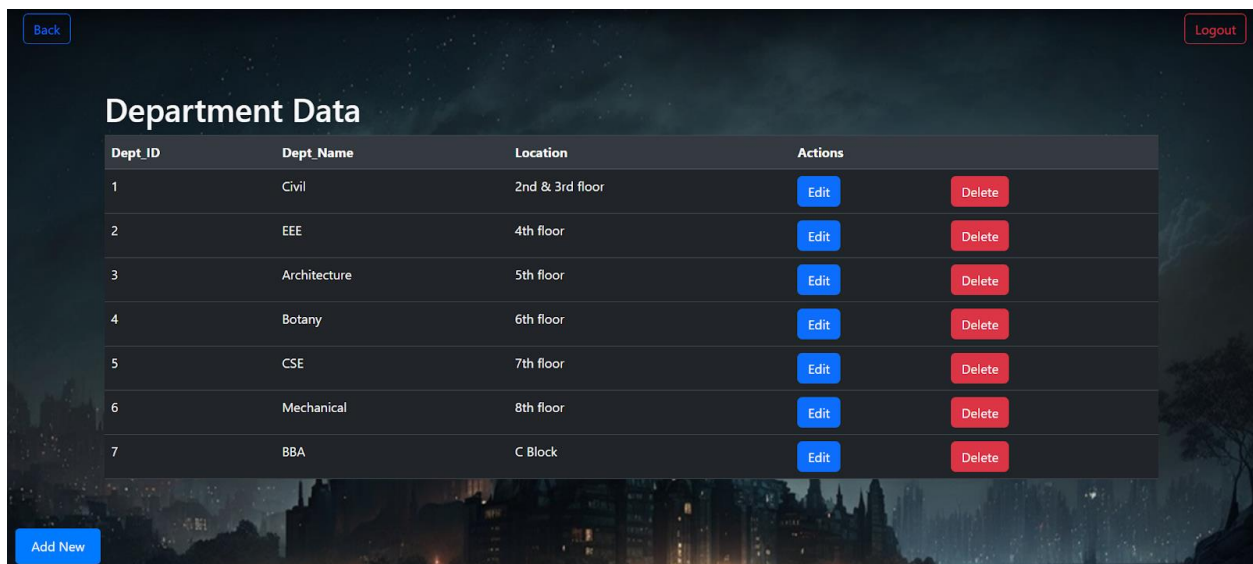
ADD DEPARTMENT INFO

Dept. Name: *

Dept. Location: *

Figure: Dept. Add Page

DeptData provides a detailed list of all departments, allowing easy access to update or view their data.



Back Logout

Department Data

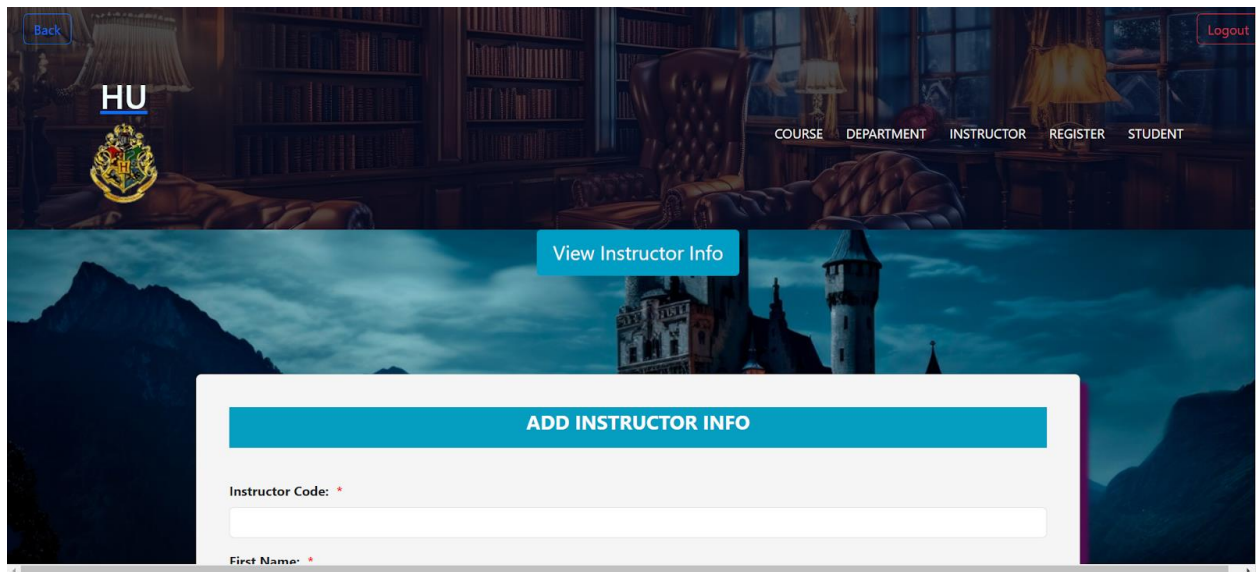
Dept_ID	Dept_Name	Location	Actions
1	Civil	2nd & 3rd floor	Edit Delete
2	EEE	4th floor	Edit Delete
3	Architecture	5th floor	Edit Delete
4	Botany	6th floor	Edit Delete
5	CSE	7th floor	Edit Delete
6	Mechanical	8th floor	Edit Delete
7	BBA	C Block	Edit Delete

Add New

Figure: Dept. Data Page

Instructor Management (InstructorAdd, InstructorData):

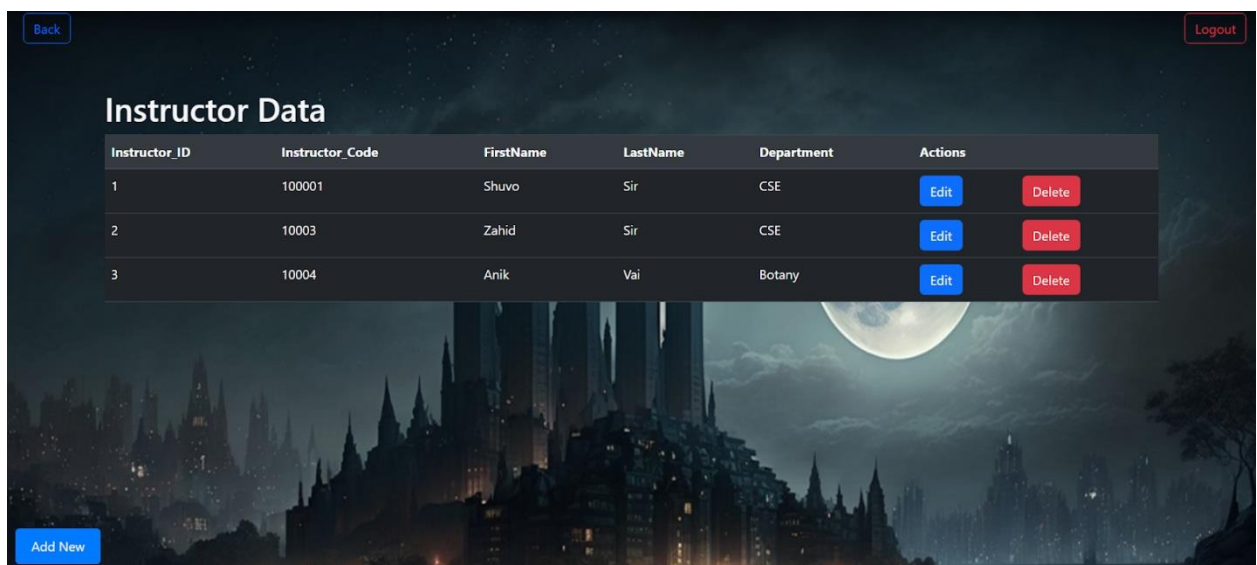
Using InstructorAdd, administrators can register new instructors, capturing their personal and professional details.



The screenshot shows the 'Instructor Add Page' of a web application. The background is a dark, atmospheric image of a castle at night. In the top left corner, there is a 'Back' button. In the top right corner, there is a 'Logout' button. Below the top navigation bar, there is a horizontal menu with the following items: COURSE, DEPARTMENT, INSTRUCTOR, REGISTER, and STUDENT. A 'View Instructor Info' button is visible in the center. The main content area features a white form titled 'ADD INSTRUCTOR INFO'. The form has two input fields: 'Instructor Code' with a red asterisk indicating it is required, and 'First Name' also with a red asterisk. The form is partially obscured by a blue banner at the top that reads 'ADD INSTRUCTOR INFO'.

Figure: Instructor Add Page

InstructorData provides a database view of all instructors, supporting the management of instructor assignments to courses and departments.



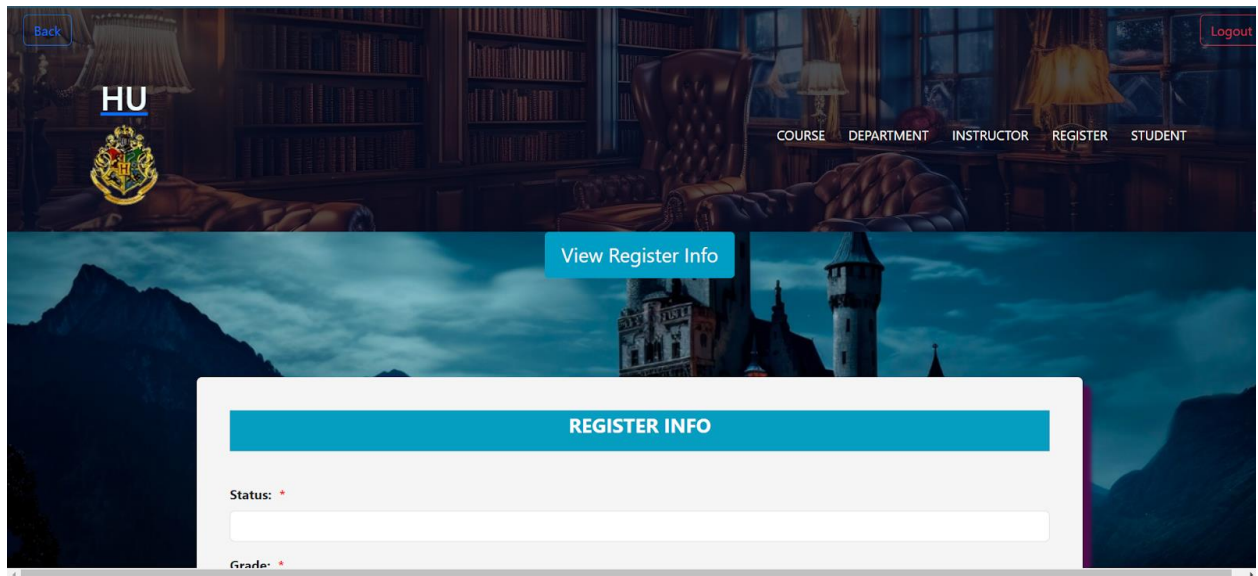
The screenshot shows the 'Instructor Data' page of a web application. The background is a dark, atmospheric image of a castle at night. In the top left corner, there is a 'Back' button. In the top right corner, there is a 'Logout' button. The page title is 'Instructor Data'. Below the title is a table with the following columns: Instructor_ID, Instructor_Code, FirstName, LastName, Department, and Actions. The table contains three rows of data. The 'Actions' column for each row contains two buttons: 'Edit' (blue) and 'Delete' (red). At the bottom left of the page, there is an 'Add New' button.

Instructor_ID	Instructor_Code	FirstName	LastName	Department	Actions
1	100001	Shuvo	Sir	CSE	Edit Delete
2	10003	Zahid	Sir	CSE	Edit Delete
3	10004	Anik	Vai	Botany	Edit Delete

Figure: Instructor Data Page

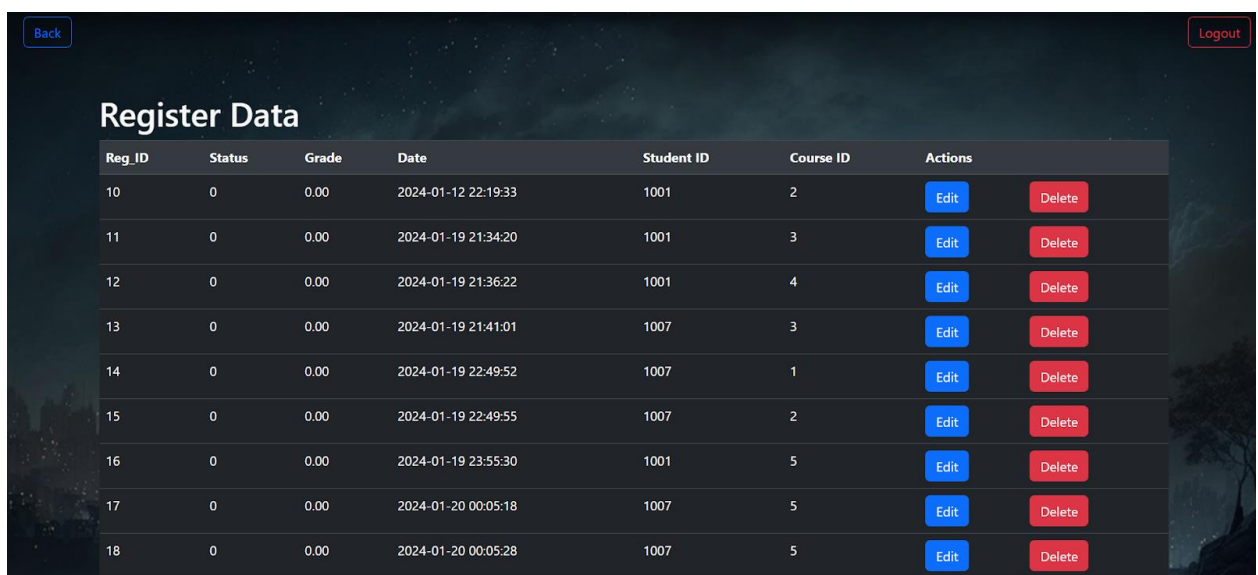
Registration System (RegisterAdd):

This module allows students to register for courses. Administrators can oversee and manage the registration process, ensuring proper course allocation based on student eligibility and course availability.



The screenshot shows the 'Register Add Page' of the HU Registration System. The page features a dark-themed header with a 'Back' button on the left and a 'Logout' button on the right. The header also includes a navigation menu with links for 'COURSE', 'DEPARTMENT', 'INSTRUCTOR', 'REGISTER', and 'STUDENT'. The main content area has a 'View Register Info' button. Below this, a 'REGISTER INFO' form is displayed, containing a 'Status' field with a red asterisk and a 'Grade' field with a red asterisk.

Figure: Register Add Page



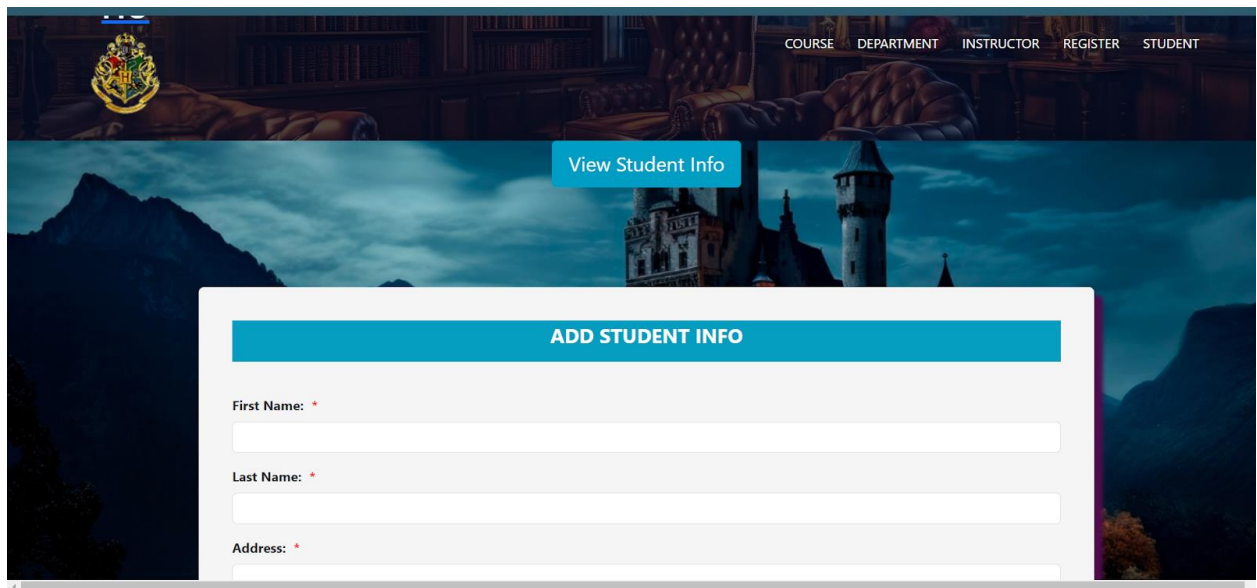
The screenshot shows the 'Register Data' page of the HU Registration System. The page features a dark-themed header with a 'Back' button on the left and a 'Logout' button on the right. The main content area displays a table with the following data:

Reg_ID	Status	Grade	Date	Student ID	Course ID	Actions
10	0	0.00	2024-01-12 22:19:33	1001	2	Edit Delete
11	0	0.00	2024-01-19 21:34:20	1001	3	Edit Delete
12	0	0.00	2024-01-19 21:36:22	1001	4	Edit Delete
13	0	0.00	2024-01-19 21:41:01	1007	3	Edit Delete
14	0	0.00	2024-01-19 22:49:52	1007	1	Edit Delete
15	0	0.00	2024-01-19 22:49:55	1007	2	Edit Delete
16	0	0.00	2024-01-19 23:55:30	1001	5	Edit Delete
17	0	0.00	2024-01-20 00:05:18	1007	5	Edit Delete
18	0	0.00	2024-01-20 00:05:28	1007	5	Edit Delete

Figure: Register Data Page

Student Management (StudentAdd, StudentData):

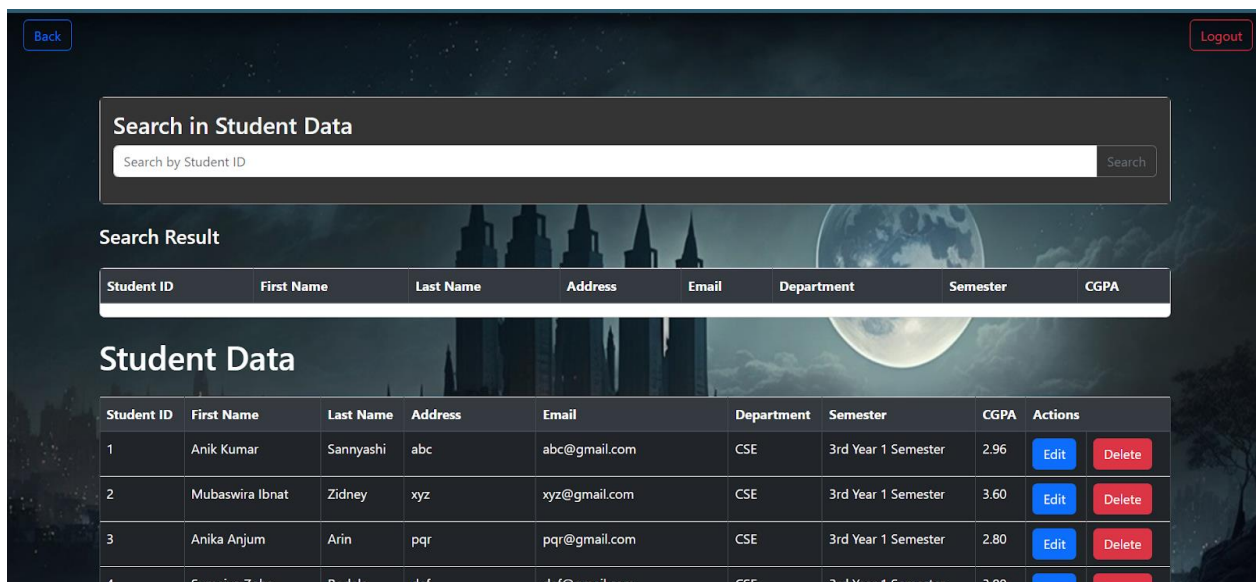
The StudentAdd feature is used to add students to the system, collecting data such as name, ID, and enrolled courses.



The screenshot shows a web application interface for adding a student. At the top, there is a navigation bar with links: COURSE, DEPARTMENT, INSTRUCTOR, REGISTER, and STUDENT. Below the navigation bar is a header image of a library. A blue button labeled "View Student Info" is visible. The main content area features a white form titled "ADD STUDENT INFO" with three input fields: "First Name:", "Last Name:", and "Address:". Each field has a red asterisk indicating it is required.

Figure: Student Add Page

StudentData serves as a student directory, with options to modify or analyze their academic performance.



The screenshot shows a web application interface for viewing student data. At the top, there is a navigation bar with links: Back, Logout, and a search bar. The search bar is labeled "Search in Student Data" and has a "Search" button. Below the search bar is a "Search Result" section. The main content area features a table titled "Student Data" with columns: Student ID, First Name, Last Name, Address, Email, Department, Semester, CGPA, and Actions. The table contains four rows of student data.

Student ID	First Name	Last Name	Address	Email	Department	Semester	CGPA	Actions
1	Anik Kumar	Sanmyashi	abc	abc@gmail.com	CSE	3rd Year 1 Semester	2.96	Edit Delete
2	Mubaswira Ibnat	Zidney	xyz	xyz@gmail.com	CSE	3rd Year 1 Semester	3.60	Edit Delete
3	Anika Anjum	Arin	pqr	pqr@gmail.com	CSE	3rd Year 1 Semester	2.80	Edit Delete
4	Sumaiya Zoha	Rodela	def	def@gmail.com	CSE	3rd Year 1 Semester	3.00	Edit Delete

Figure: Student Data Page

Student Login

This page allows students to log into the system using their credentials.

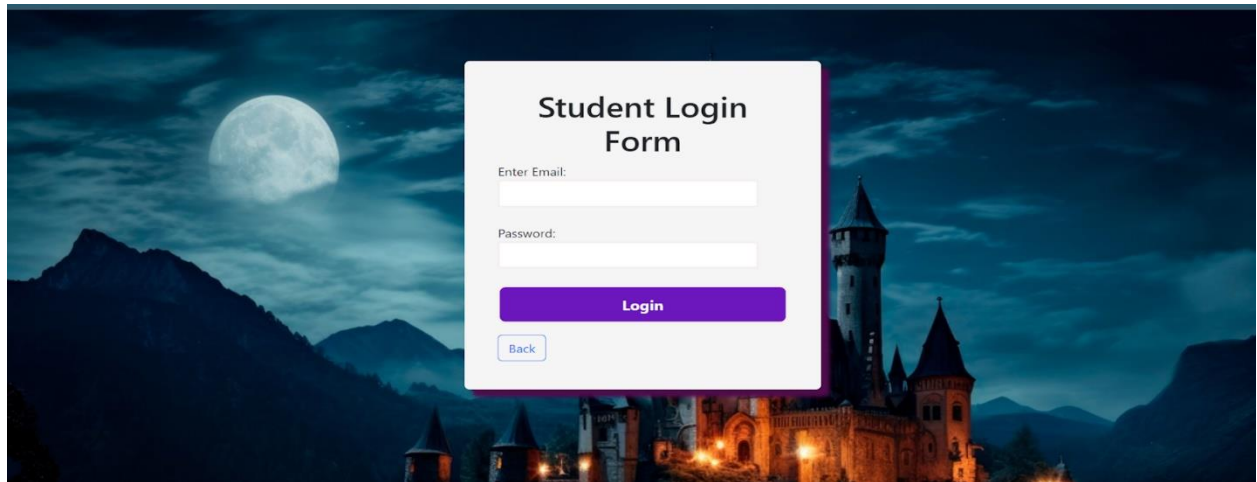


Figure: Student Login Page

- Username/Email Field: A text input where the student enters their username or registered email.
- Password Field: A password input field masked with asterisks for privacy.
- Login Button: A clickable button to submit the login form, initiating authentication.

Course Search for Students

This page allows students to search for courses they can register for in the current or upcoming semester.

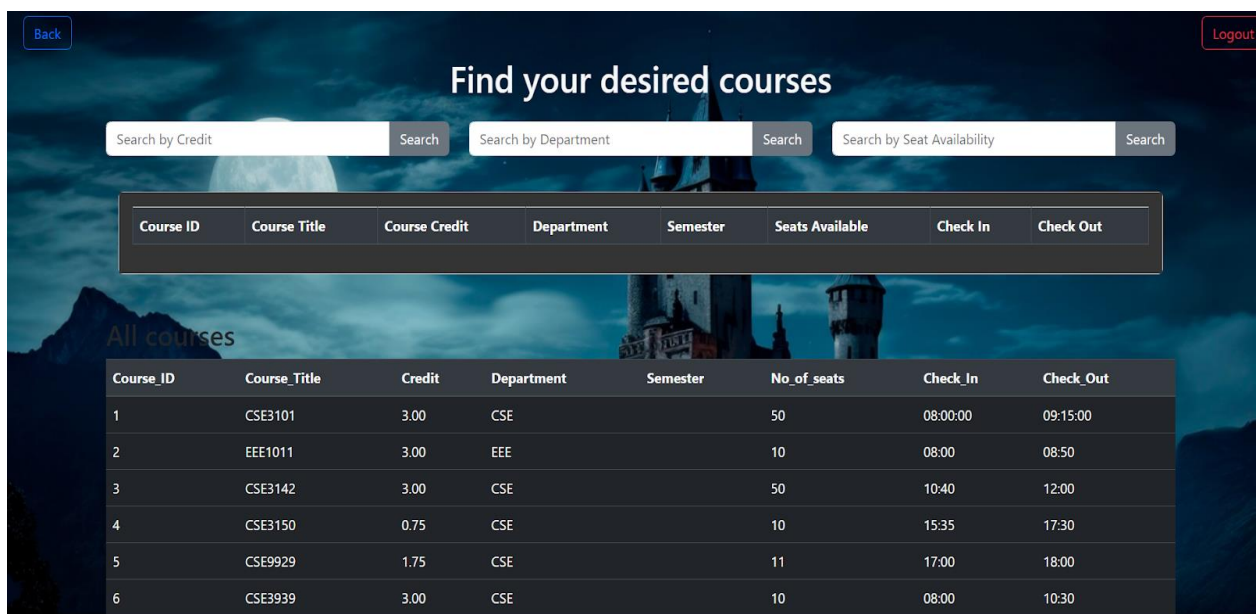


Figure: View Course Page

- Search Bar: A text input field where the student can search for courses by name, code, or instructor.

- Filter Options: Optional dropdowns or filters to refine searches by department, course type, or schedule.
- Search Button: A button to execute the search based on the criteria entered in the search bar or filters.
- Search Results Section: A dynamic section where the list of available courses is displayed, with details such as course name, instructor, timings, and availability.

Register for Course

This page provides students with the option to register for a selected course after viewing its details.

REGISTER FOR COURSE

Course ID:
CSE3101 (1)
SUBMIT

Search for courses
Search by Credit, Department or Title Search

Search result Back Logout

Course ID	Course Title	Course Credit	Department	Semester	Seats Available	Check In	Check Out
-----------	--------------	---------------	------------	----------	-----------------	----------	-----------

Figure: Register for Course Page

- Course Details: The selected course's name, code, instructor, schedule, and available seats.
- Confirm Registration Button: A large button that confirms the student's enrollment into the course.
- Back to Search Button: A secondary button allowing students to return to the search page to choose other courses.

Register Data

This page provides students with a summary of their registered courses, often displayed after they've completed the registration process.

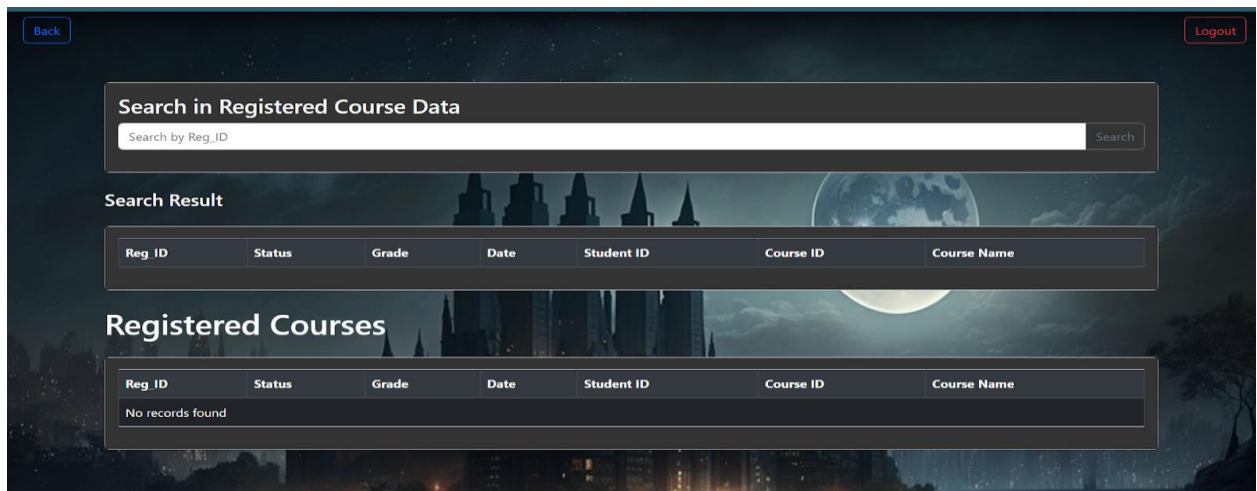
The screenshot shows a web interface for 'Register Data'. At the top left is a 'Back' button and at the top right is a 'Logout' button. Below these is a search bar titled 'Search in Registered Course Data' with a placeholder 'Search by Reg_ID' and a 'Search' button. Under the search bar is a 'Search Result' section. Below that is a table titled 'Registered Courses'. The table has columns: Reg_ID, Status, Grade, Date, Student ID, Course ID, and Course Name. The table content shows 'No records found'.

Figure: Register data for student Page

- **Course List:** Displays the list of courses the student has successfully registered for, including details like course name, schedule, and instructor.
- **Total Credits/Hours:** A section showing the total number of credits or hours registered, helping students stay within their academic limits.

Student Profile

This page allows students to view their personal information, such as contact details, and review their academic progress.

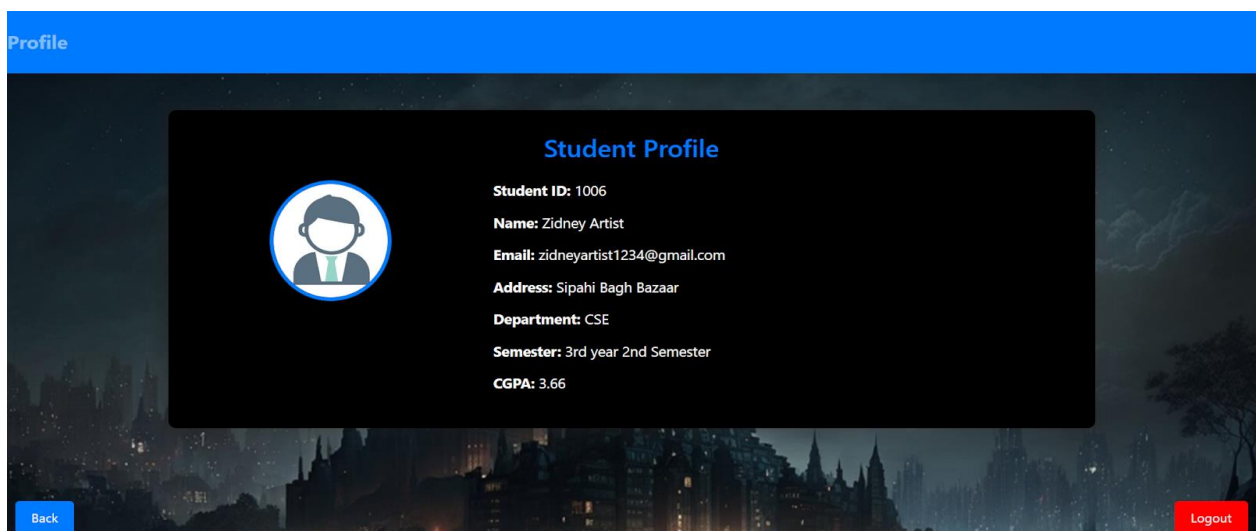
The screenshot shows a 'Student Profile' page. At the top left is a 'Back' button and at the top right is a 'Logout' button. The page has a blue header with the word 'Profile'. The main content area is dark and contains a circular profile picture placeholder on the left. To the right of the picture is the title 'Student Profile' in blue. Below the title are several lines of text: 'Student ID: 1006', 'Name: Zidney Artist', 'Email: zidneyartist1234@gmail.com', 'Address: Sipahi Bagh Bazaar', 'Department: CSE', 'Semester: 3rd year 2nd Semester', and 'CGPA: 3.66'.

Figure: Student Profile Page

Opportunities of Development

There are numerous opportunities for future improvement and development with the university course registration project. Enhancing the existing features will allow us to integrate cutting-edge technologies, optimize workflows, and enhance user experience. These are a few significant opportunities:

1. User Experience Enhancement:

- **Personalized Dashboards:** Develop personalized user dashboards that offer recommendations based on previous selections, academic performance, and career goals.
- **Mobile Application:** Create a mobile app to enable students to manage their registrations and schedules on the go, increasing accessibility and convenience.

2. Advanced Search Capabilities:

- **AI-Powered Search:** Implement machine learning algorithms to improve the course search functionality, allowing for natural language processing to better understand user queries.
- **Filters and Sorting Options:** Enhance search options with advanced filters (e.g., course difficulty, professor ratings, prerequisites) to help users find courses that best fit their needs.

3. Integration with Other Systems:

- **Learning Management Systems (LMS):** Integrate with existing LMS platforms to provide seamless access to course materials and schedules.
- **Advising Tools:** Collaborate with academic advising tools to provide students with real-time advice based on their academic progress and future career paths.

4. Data Analytics and Reporting:

- **Analytics Dashboard:** Develop an analytics dashboard for administrators to track course registration trends, student preferences, and resource allocation.
- **Predictive Analytics:** Use predictive analytics to identify at-risk students and provide timely interventions to improve retention rates.

5. Enhanced Scheduling Features:

- **Conflict Resolution:** Implement an intelligent scheduling tool that automatically detects and resolves time conflicts in course selections.
- **Real-Time Availability:** Allow real-time updates on course availability, ensuring students have the most accurate information when making selections.

6. Collaborative Features:

- **Peer Reviews and Ratings:** Introduce a system for students to rate and review courses, fostering a community-driven approach to course selection.

7. Feedback Mechanism:

- **User Feedback Loop:** Establish a feedback mechanism that allows users to provide suggestions and report issues, driving continuous improvement.

By exploring these opportunities, the university course registration project can evolve into a comprehensive platform that not only meets current needs but also adapts to future educational landscapes. This focus on continuous improvement and innovation will enhance student satisfaction and streamline academic processes.

Contribution

20210104077

- Home Page
- About Page
- Department Page
- Signup Page
- Login Page
- Logout Page
- Instructor Page
- Student Signup Page
- Course Data Page
- Edit Course Page
- Edit Department Page
- Edit Instructor Page
- Edit Registration Page
- Edit Student Page
- Delete Course Page
- Delete Department Page
- Update Course Page
- Update Department Page
- Update Instructor Page
- Student login
- Register and Registered Courses

20210104098

- Admission Page
- Contact Page
- Admin Page
- Course Page
- Department Data Page
- Instructor Data Page
- Registration Data Page
- Student Data Page
- Delete Instructor Page
- Delete Registration Page
- Delete Student Page
- Update Registration Page
- Update Student Page
- Student Dashboard & Student Profile
- View Courses
- All CSS files (Admin CSS, Style C CSS, Style Z CSS)

Workload Distribution (Using pie chart):

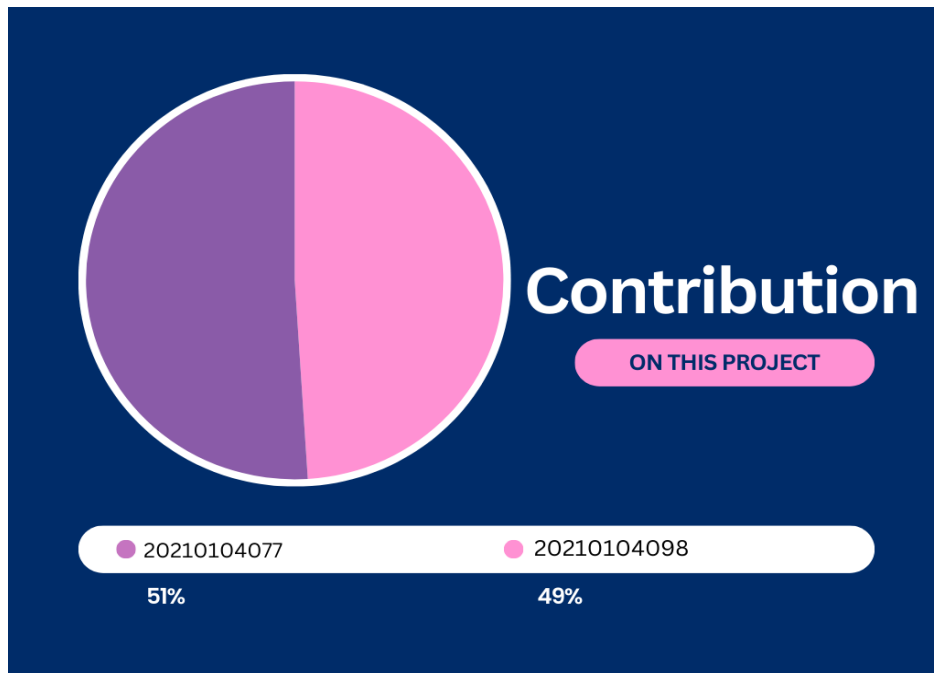


Figure 18: Workload Distribution

Conclusion

Our university course registration system combines front-end and back-end technology to produce an effective, user-friendly platform. After conducting a thorough analysis of open credit systems and utilizing various tools including PHP, MySQL, and Bootstrap, we have developed a system that effectively simplifies the course registration procedure while providing flexibility and user-friendliness. The project guarantees an improved experience for administrators and students alike by satisfying the specifications of a contemporary, scalable, and secure web application.

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